

Week2

Robot Arm Kinematic Model

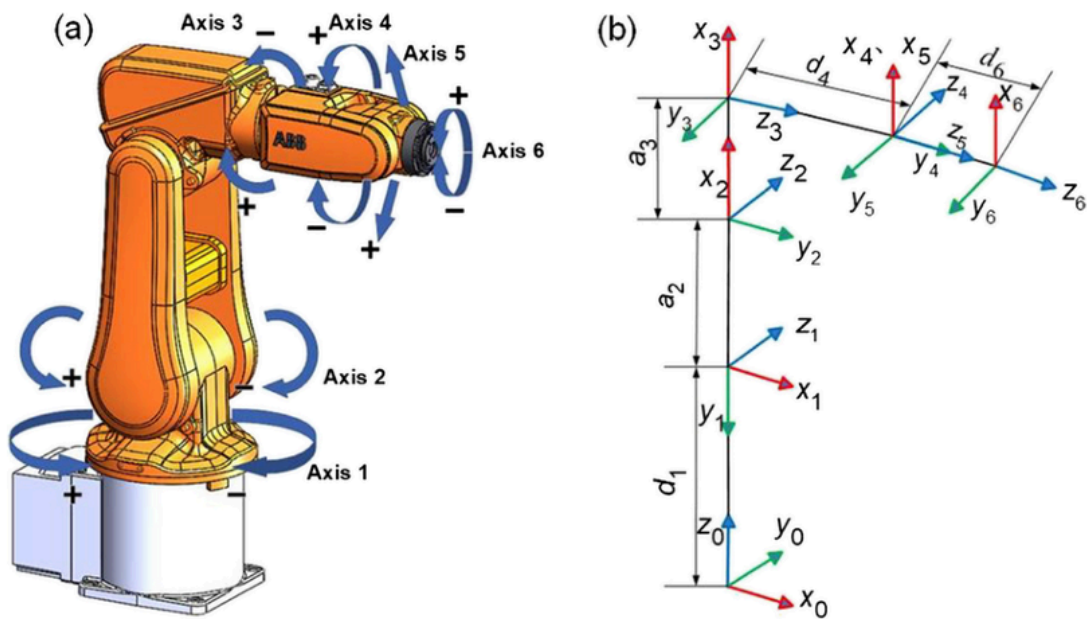


Figure 1.1: IRB120 mechanical model(a) and kinematic model(b)

Mechanical joints used in robotics

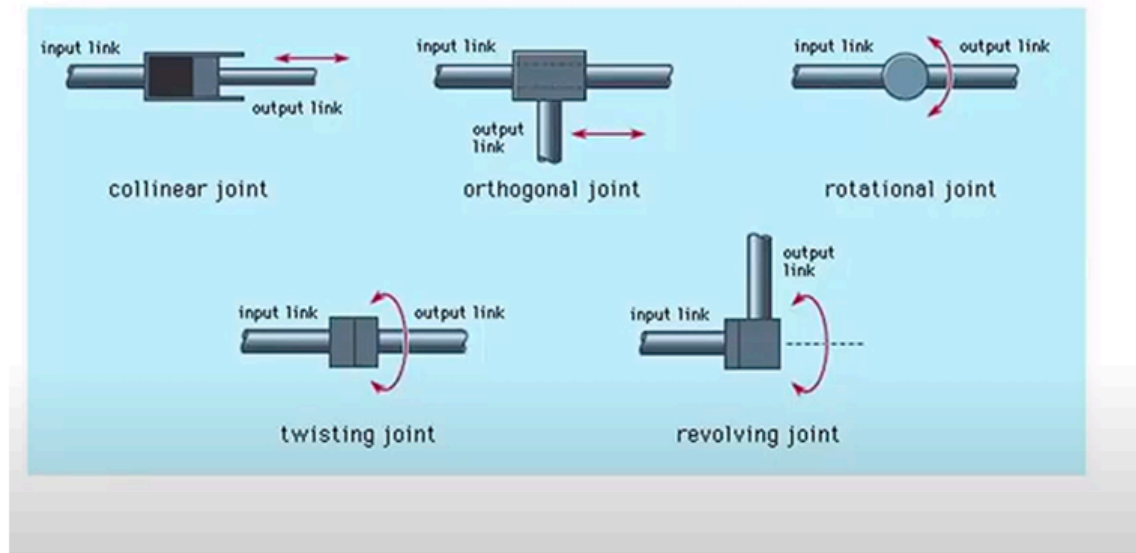


Figure 1.2: Joint types

This robot has two basic components:

- The **joints**, also known as **axes**, are the **movable parts** of the robotic arm. One of the common joints is the **Revolute Joint**, which rotates around its Z axis
- The **Links**: the rigid structural components of the robotic arm that hold all the parts of the robot together; usually, they **connect between joints**.

Activity One:

From [Figure 1.1](#) (a), (b) and [Figure 1.2](#), list all the links, joints, and joint type in the order from the base to Axis 6 of the robot.

Answer:

The robot has a total of 6 parts, excluding the base on which it is fixed, but it can rotate around the Z-axis.

Link/Joint Description	Joint Type
Base -Axis1	Twisting Joint

Axis1-Axist 2	Rotational Joint
Axis 2 - Axis 3	Rotational Joint
Axis 3 - Axis 4	Twisting Joint
Axis 4 - Axis 5	Rotational Joint
Axis 5 - Axis 6	Twisting Joint

Robot Arm Simulation

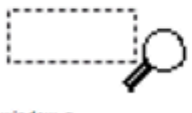
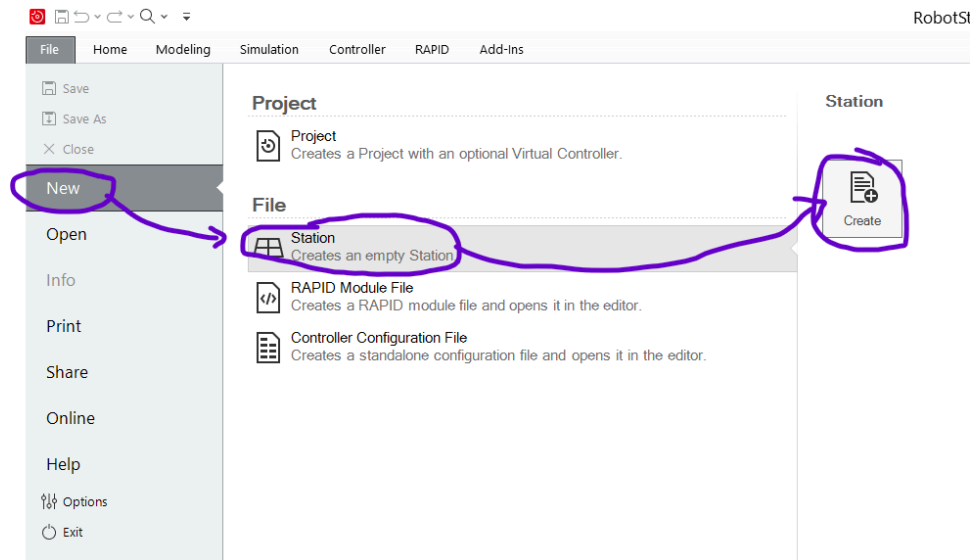
To	Use the keyboard /mouse combination	Description
Select items  selectio	 left-cl	Just click the item to select. To select multiple items, press CTRL key while clicking new items.
Rotate the station  rotate	CTRL + SHIFT +  left-cl	Press CTRL + SHIFT + the left mouse button while dragging the mouse to rotate the station. With a 3-button mouse you can use the middle and right buttons, instead of the keyboard combination.
Pan the station  pan	CTRL +  left-cl	Press CTRL + the left mouse button while dragging the mouse to pan the station.
Zoom the station  zoom	CTRL +  right-cl	Press CTRL + the right mouse button while dragging the mouse to the left to zoom out. Dragging to the right zooms in. With a 3-button mouse you can also use the middle button, instead of the keyboard combination.
Zoom using window  window_z	SHIFT +  right-cl	Press SHIFT + the right mouse button while dragging the mouse across the area to zoom into.
Select using window  window_s	SHIFT +  left-cl	Press SHIFT + the left mouse button while dragging the mouse across the area to select all items that match the current selection level.

Figure 2.1: navigate interface

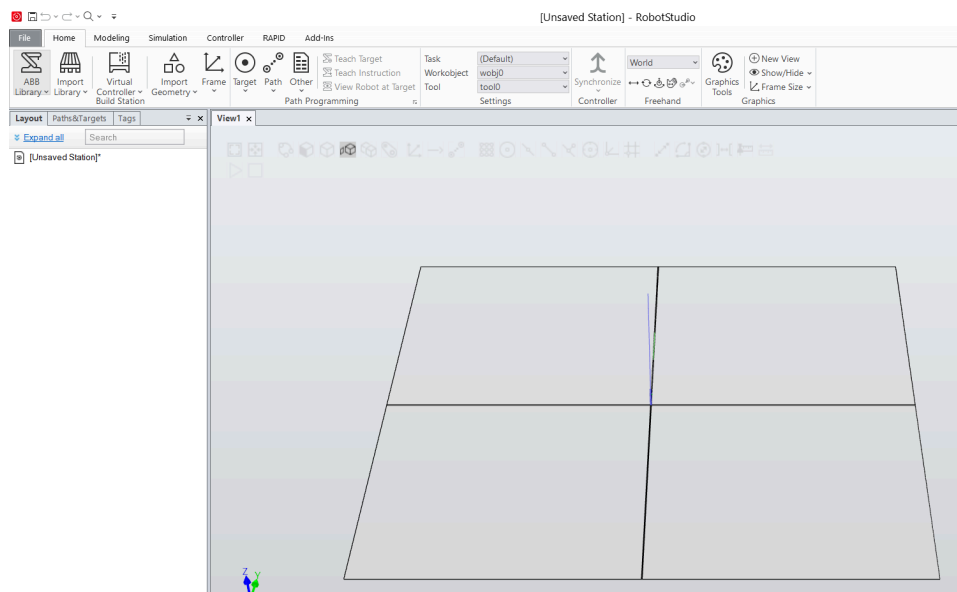
Simulation for Week 2:

Creating the file and adding a robot with a controller

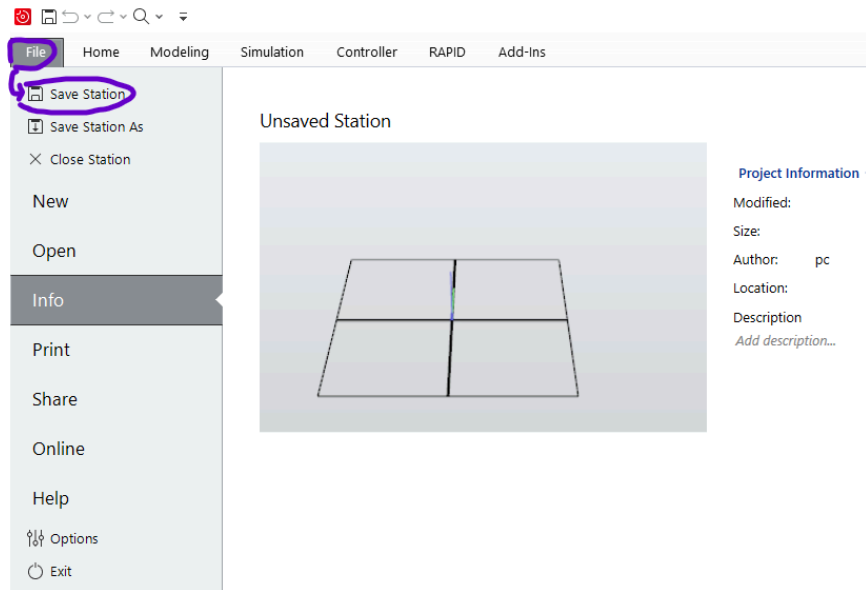
1. This will create workstation: Open RobotStudio → New → File → station → Create



2. A new station will open.

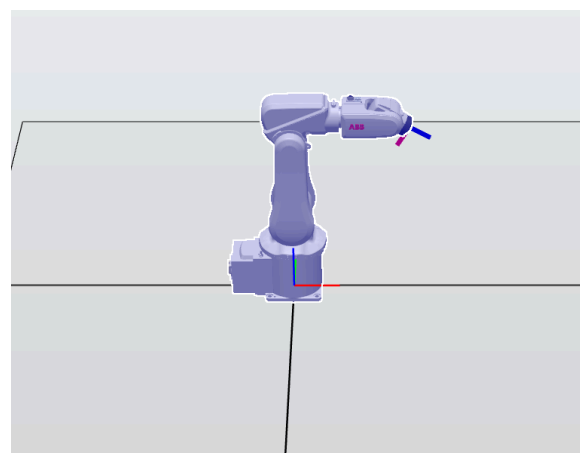
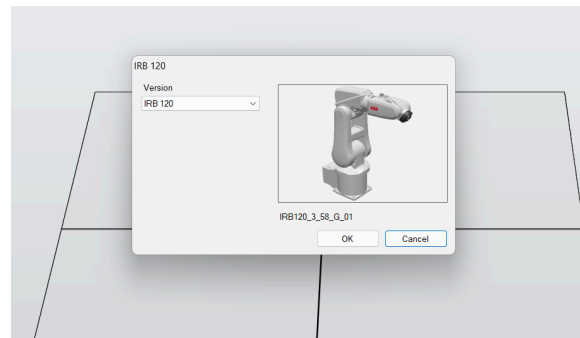
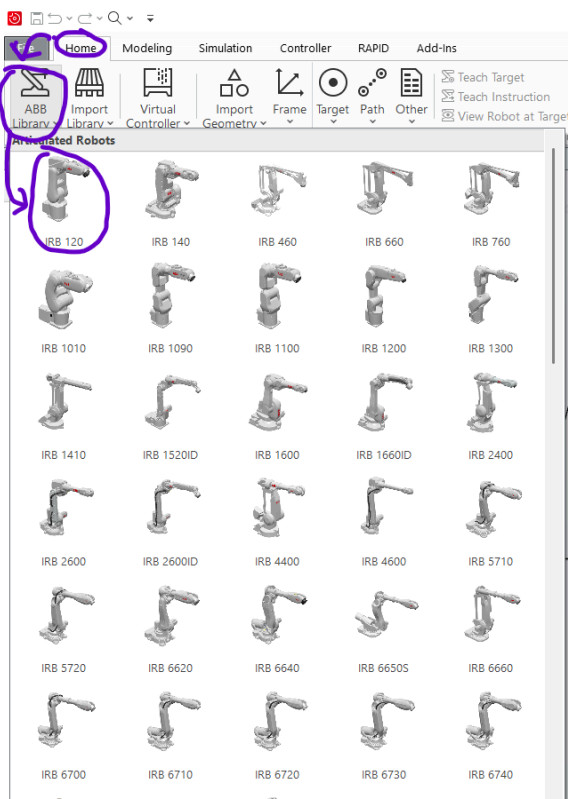


3. File → Save Station.

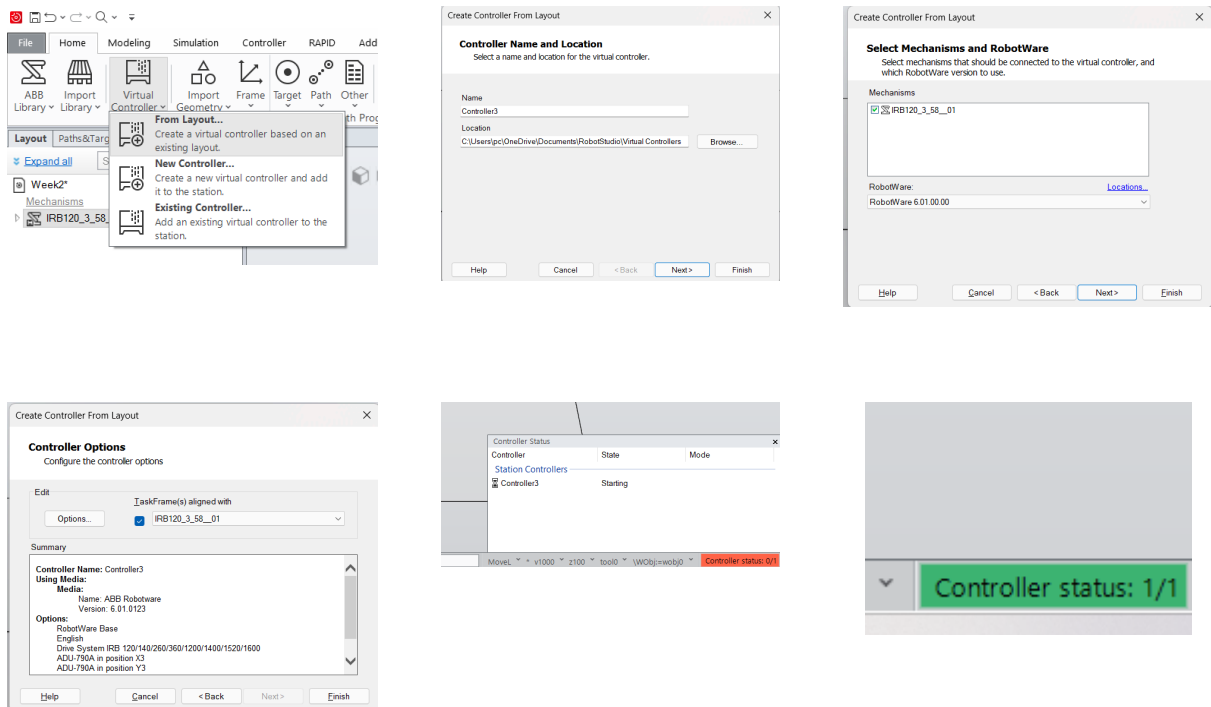


4. Save the station on a local drive in your local drive, not on one drive or iCloud.

5. Add your first robot to the station: Home → ABB Library → IRB12 Press OK.



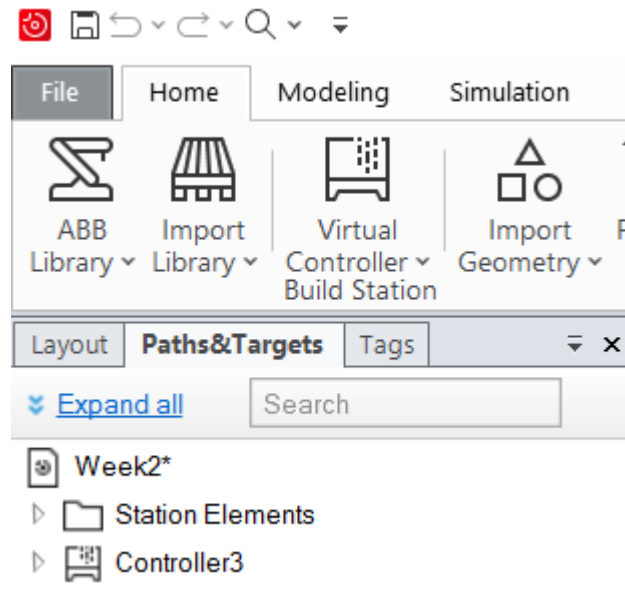
6. To control the first robot mechanism (body), add a controller: Home → Virtual controller
 → From layout → next → Select this configuration: Robot Model: IRB120 → RobotWare: 6.01 → Finish → next → finish



Note: Wait for the controller status to turn green.

If the controller status in the bottom right corner is not green, then double-check your steps.

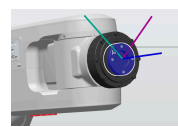
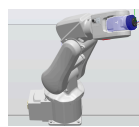
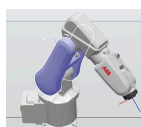
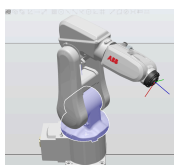
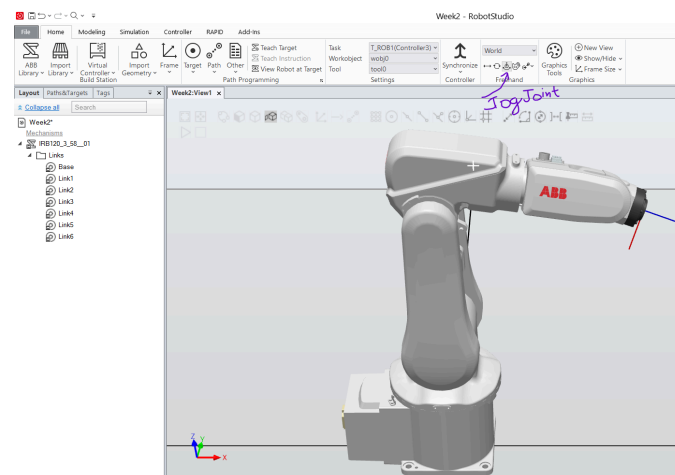
7. You should be able to see that a controller has been added in the Path&Targets tab on the left side.
 Panel



Manual control/jog robot

When using the jog Joint (next to the linear jog icon or Move icon inside the home tab (top panel)

—fig: Figure 2.4) function in the robot studio to control the robot. Firstly, try to move all the joints individually, then answer the question below with the joint jog function.



2.2.1. How many joints move at the same time?

Answer: Only the selected joint move is selected

2.2.2. Which parts of the robot can be controlled with this function?

Answer: all parts shown in the above image can be moved except the base on which the robot is mounted

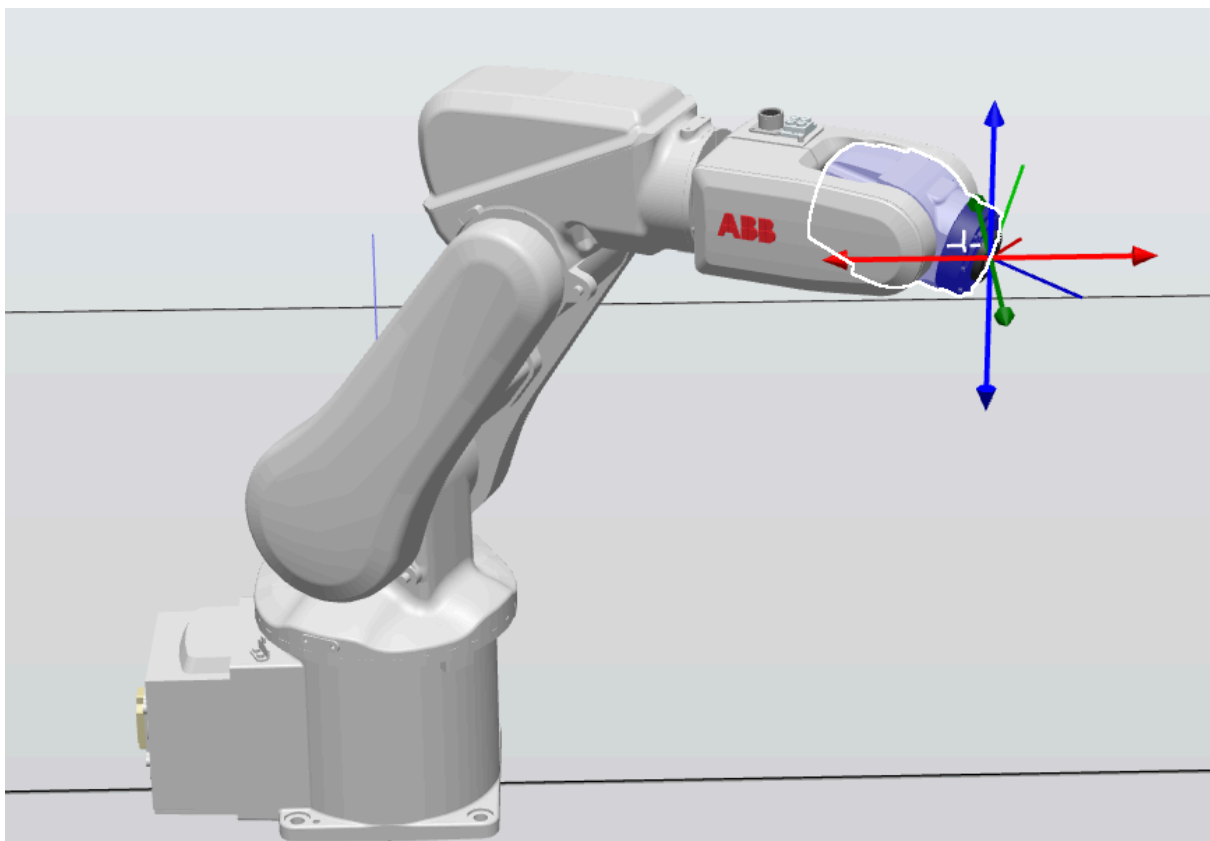
When using the jog Linear (Move icon) function in the robot studio to control the robot:

2.2.1. How many joint moves at the same time?

Answer: A jog liner moves all necessary joints that are required to move in that particular direction.

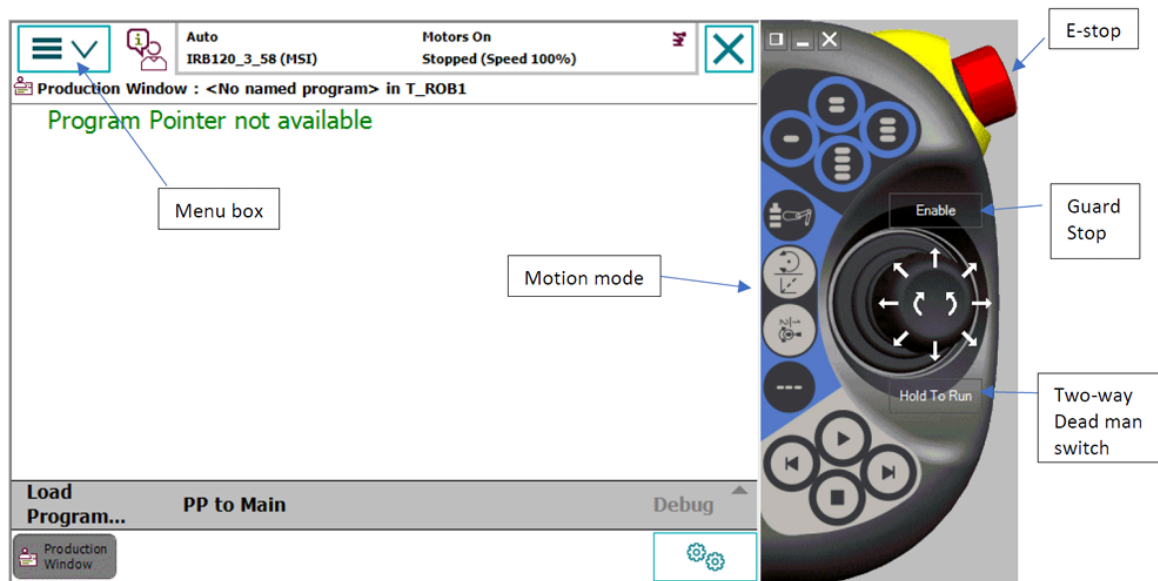
2.2.2. What part of the robot can be controlled with this function?

Answer: all parts shown in the above image can be moved except the base on which the robot is mounted

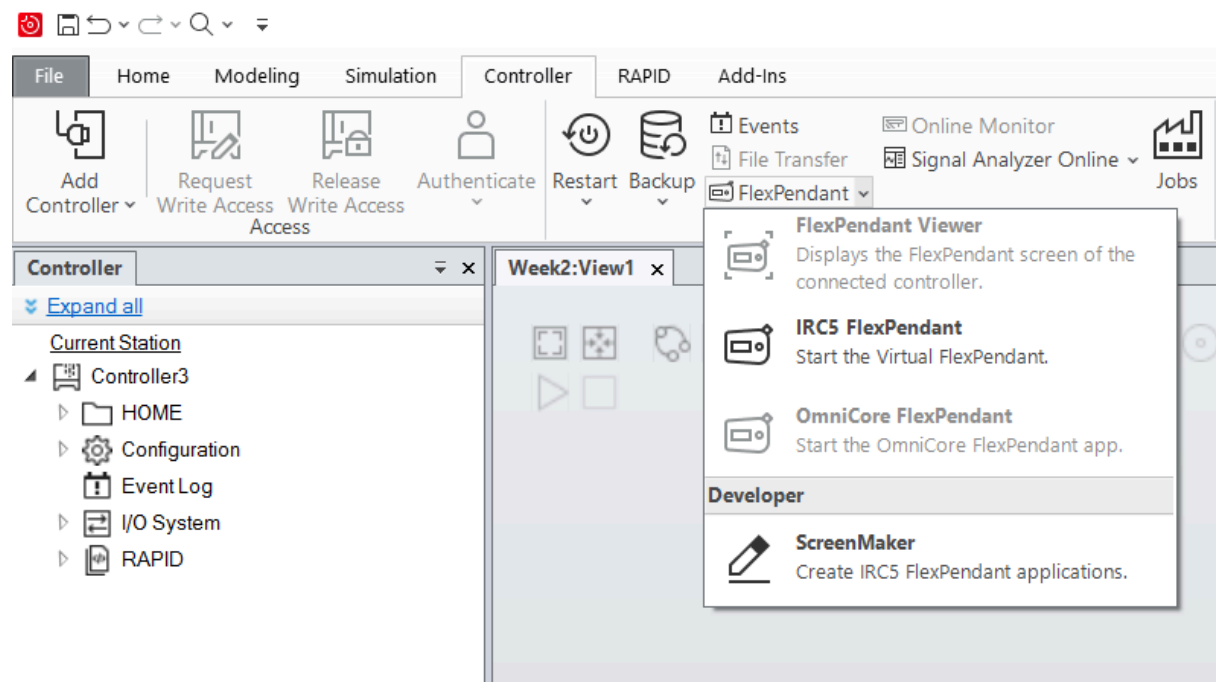


2.3 Manual control robot with the FlexPendant

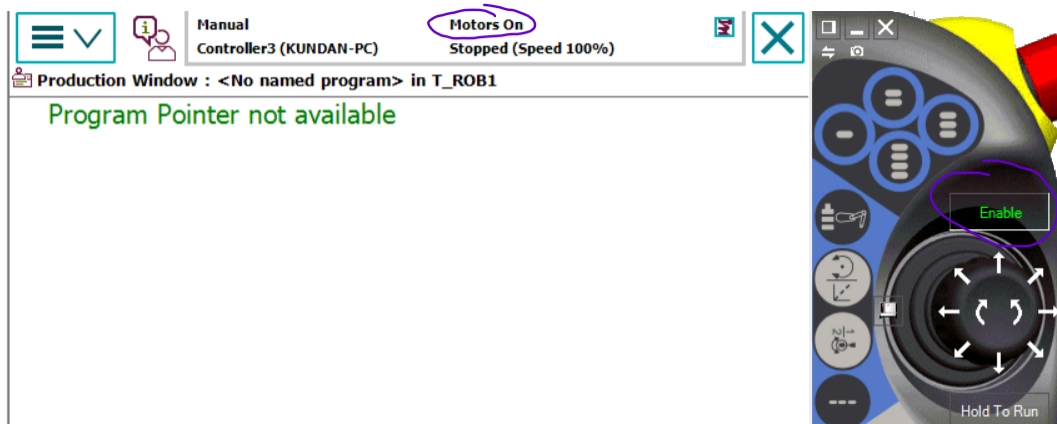
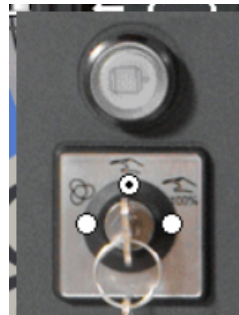
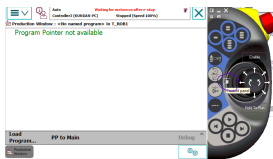
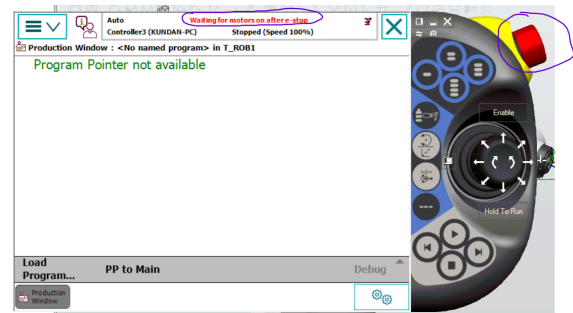
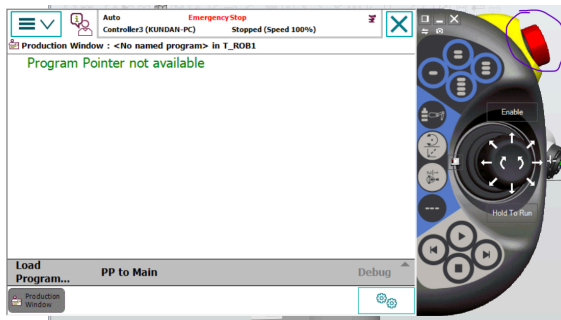
FlexPendant can be used to program, teach, monitor, and operate the robot.



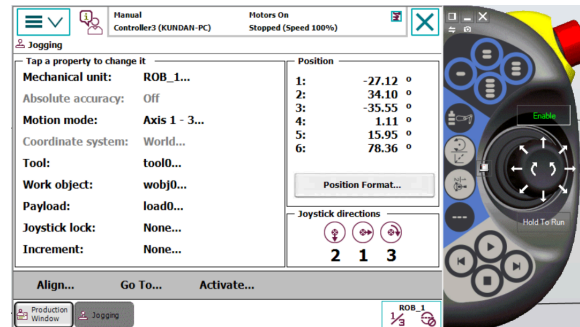
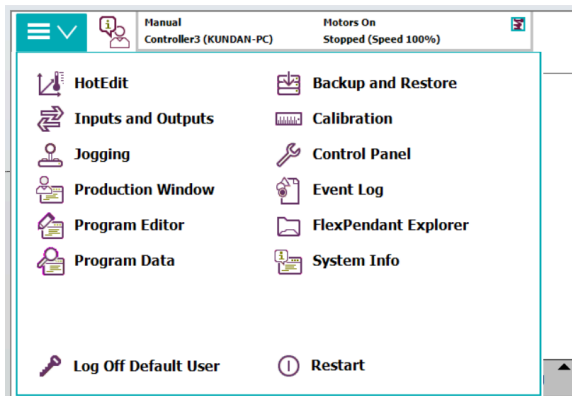
2.3.1. To open it: Controller → FlexPendant → IRC5 FlexPendant (new window pop up)



2.3.2. To release the safety mechanism: → Release E-stop → Open the controller box (click in The white box icon) → switch key to Manual mode → Press the Motors ON signal → Enable Guard stop.



2.3.3. To open the jogging monitor screen: Click menu box (Three dashes at top left) →Jogging



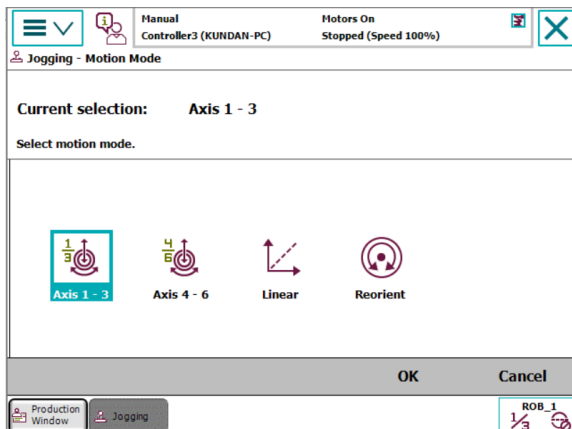
2.3.4. To change the jogging mode: Click in motion mode, → select linear or Axis option

2.3.5. To manually jog the robot: Rotate or move the thumb in the arrow direction

2.3.6. Try different motion modes to get familiar with working with the real robot

a) Axis 1-3

b) Axis 4-6 c) Linear d) Reorient.



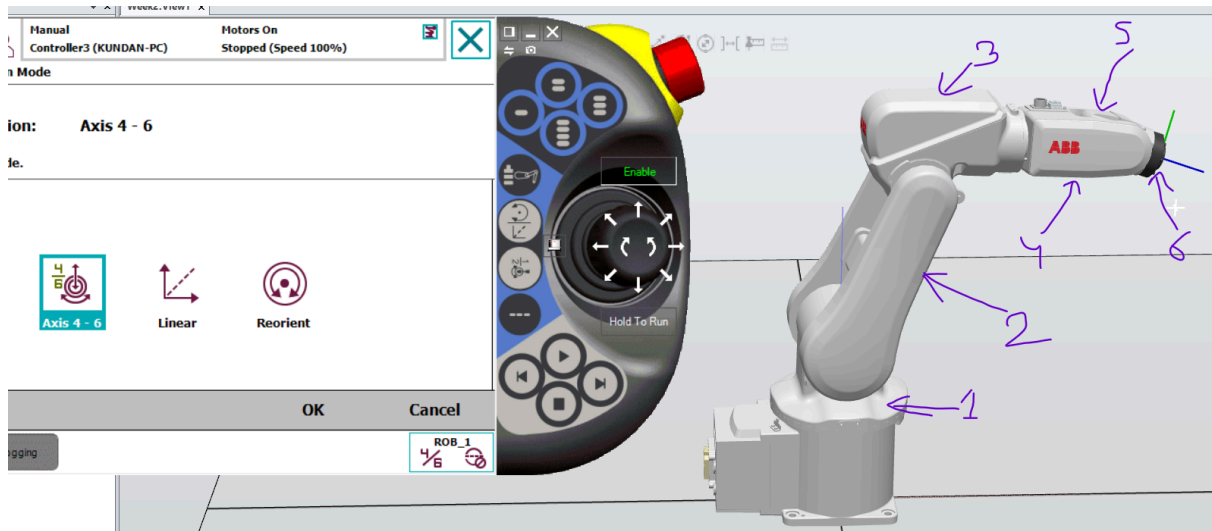
To get familiar with the station controlling system of the real robot for minor project demon

stration, write down the joystick directions and corresponding movement of the robot:

When axis 1-3 is selected:

1. The left and right arrows rotate axis 1 so that the whole robot can rotate.

2. Up and down arrow moves axis 2 so that robots move forward and backward.
3. center arrow with turn moves axis 3 so that the robot head moves up and down.



When axis 4-6 is selected:

1. Left and right arrow rotate axis 4 so that the whole robot can rotate its head where the ABB logo is present.
2. Up and down arrow rotate axis 5 so that the robot's head moves up and down.
3. Centre arrow with turn moves axis 6 so that the robot's tool head can rotate.

When Linear is selected:

1. Left and right arrow rotate its head where the ABB logo is present.
2. The up and down arrow robot moves forward and backward.
3. Centre arrow with turn head up and down.

When Reorient is selected:

Got confused what is happening with robot.

3 Working Envelope

There are many configurations of robots because they are designed for different purposes, which leads to different working envelopes. Robots can be classified based on different criteria, such as use, movement, and structure. Some common configurations of robots are Cartesian robots, SCARA robots, cylindrical robots, delta robots, and polar robots.

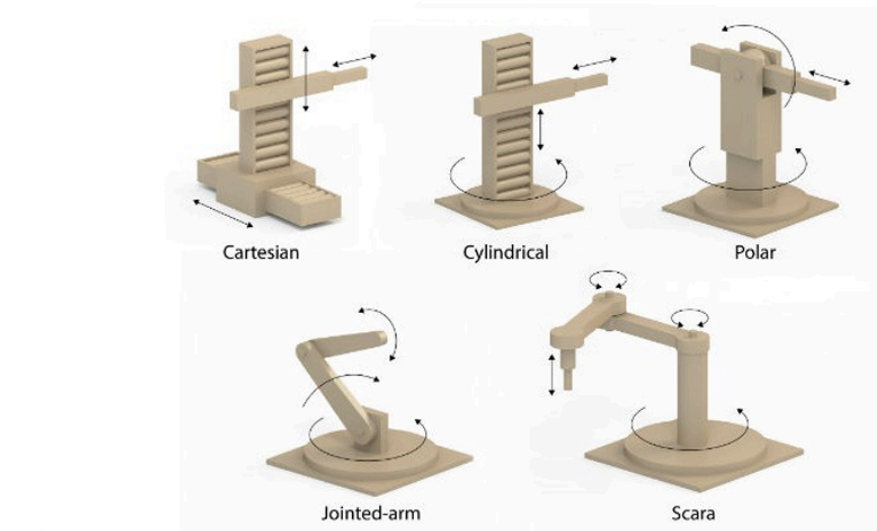


Figure 3.1: Robot configuration

Based on Figure 3.1, Figure 1.1 and your research, what is the configuration of the ABB IRB120?

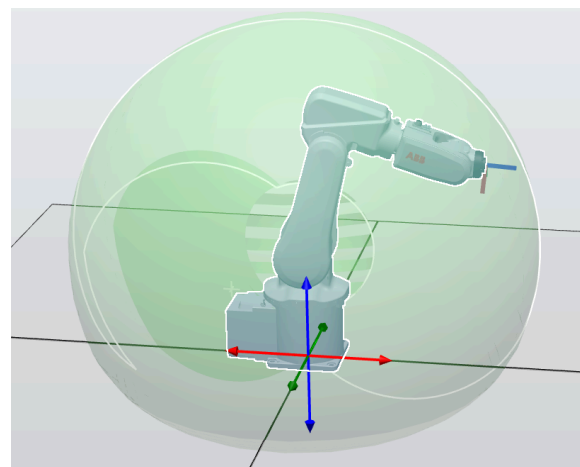
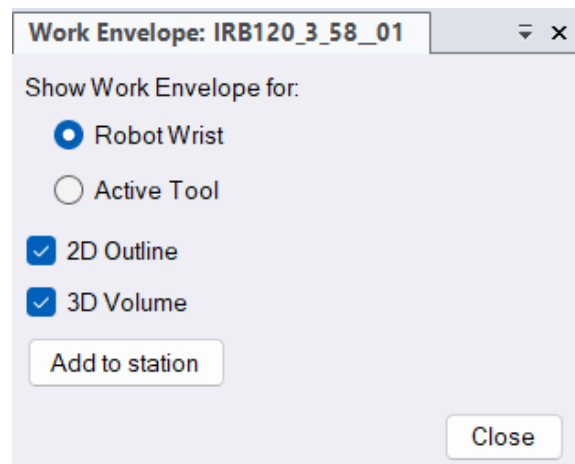
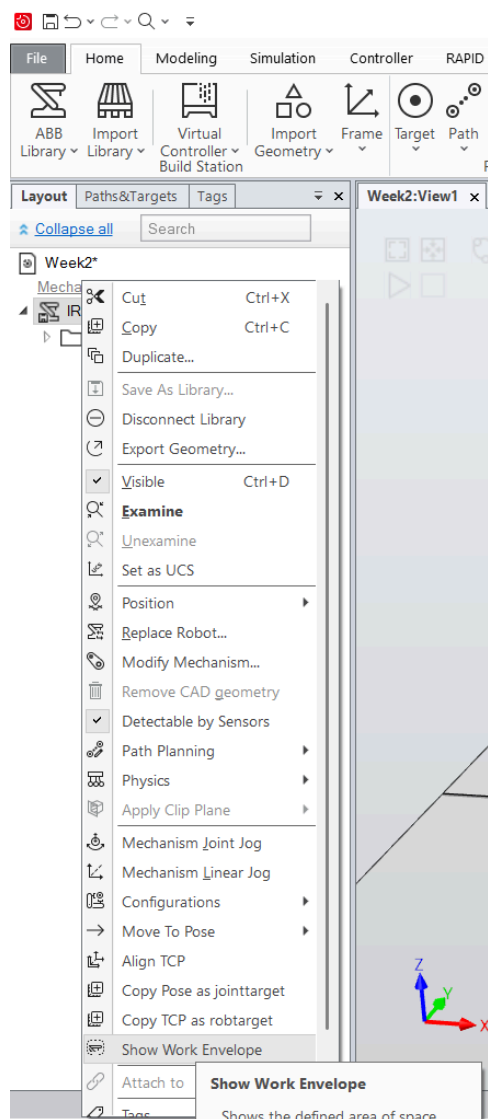
Answer: IRB120 is a jointed arm robot.

3.0.1. Go to the Home tab

3.0.2. Select the Layout tab from the left panel

3.0.3. Enable the working envelope (right-click on the robot mechanism → show work envelope

→ Tick on the 3D volume to show it in 3D- Figure 3.2) and try to move the End Effect of the robot outside the envelope.



Are you able to do it? Explain your answer below:

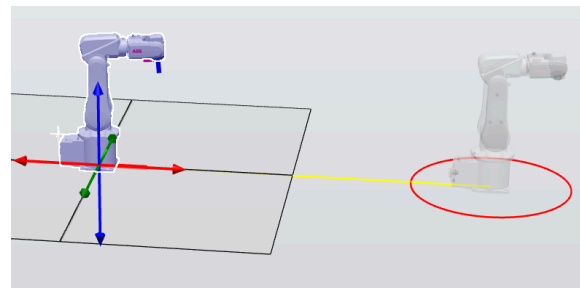
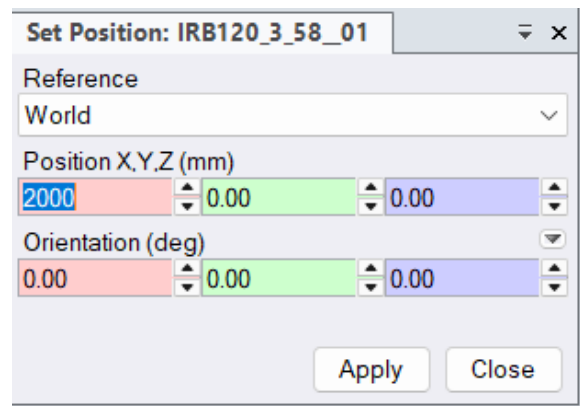
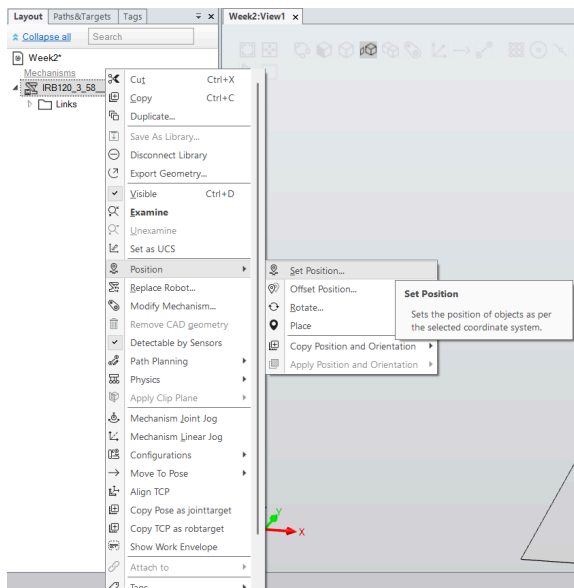
Answer: No robot is not able to move outside the working envelope.

3.1 ABB IRB1600

Move your IRB120 2m away from the centre on X direction so it doesn't overlap with the new robot using these steps:

3.1.1. Go to layout tab on left panel, right-click on the mechanism

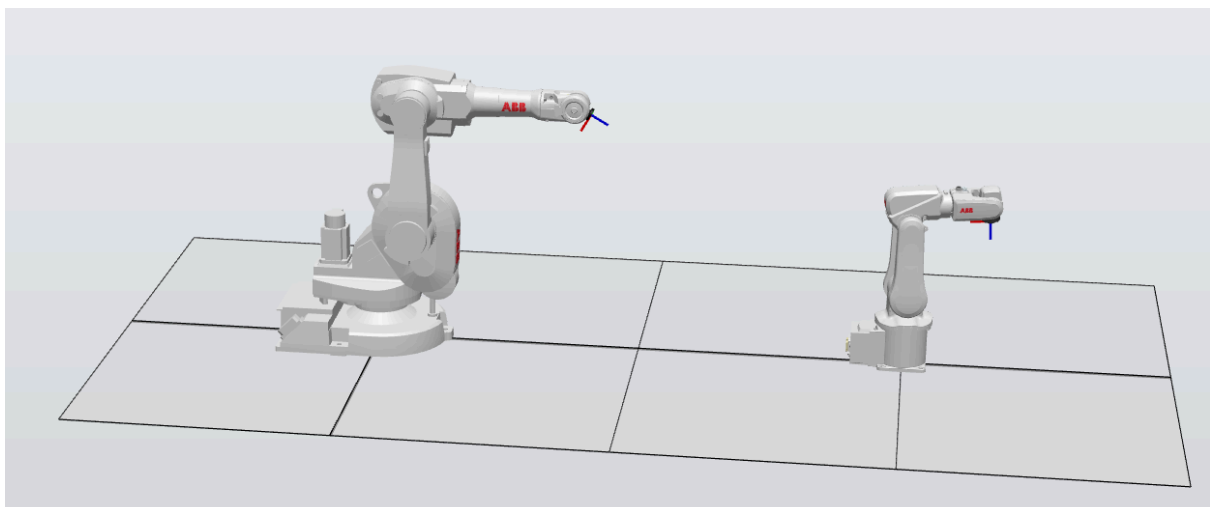
3.1.2. Select position → set position → type 2000 to Position X box (red)



- It will ask do you want to move task frame you can press No

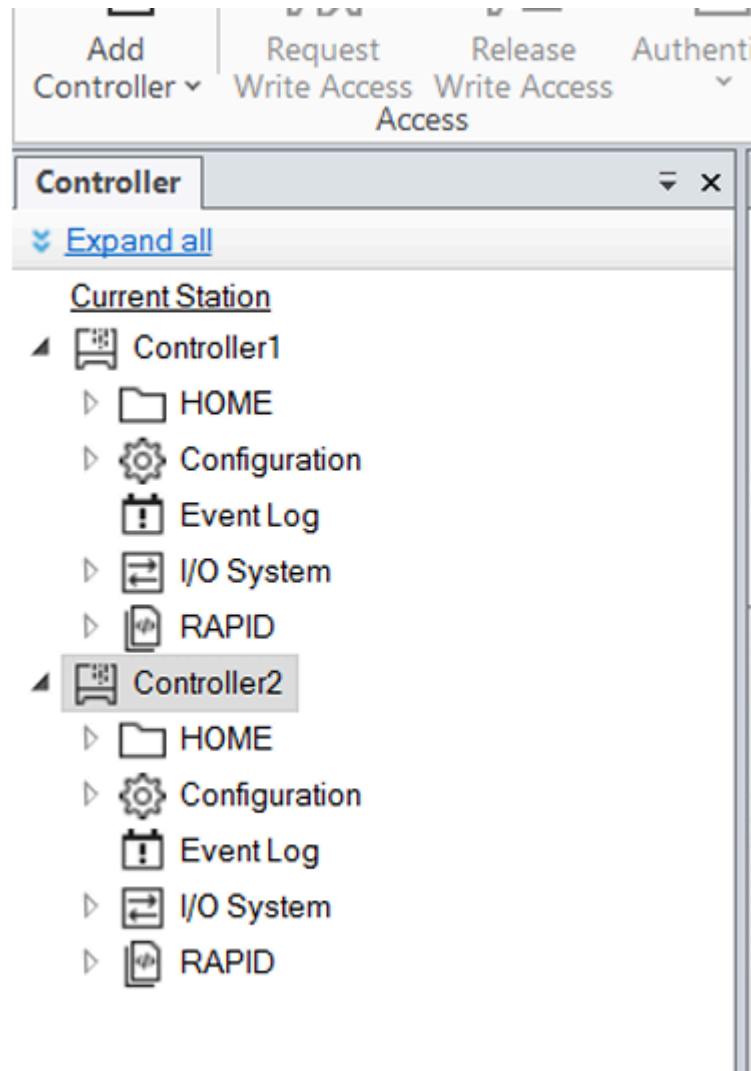
Add the IRB1600 to your workstation

Following instruction in section chapter 2 and chapter 3 to set up the controller for the new IRB1600 robot and its working envelope.

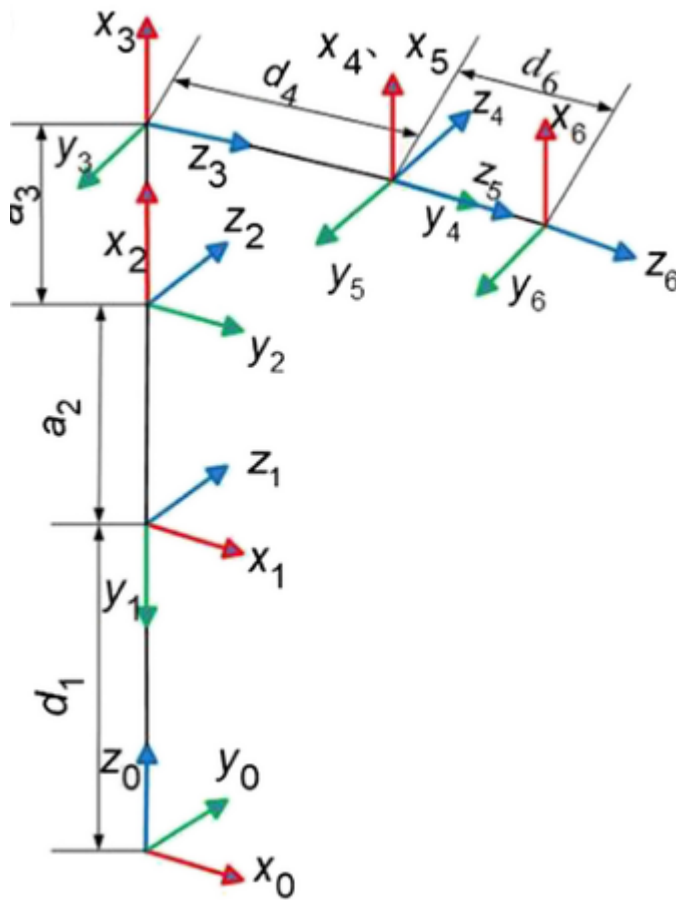


Steps to follow:

1. Add new robot
2. add new controller
3. Select the controller for robot you want to controller
4. add the flex pendant for that robot
5. if you want to change the which robot you want to control you can close the pendent and select a different controller than again add the flex pendant



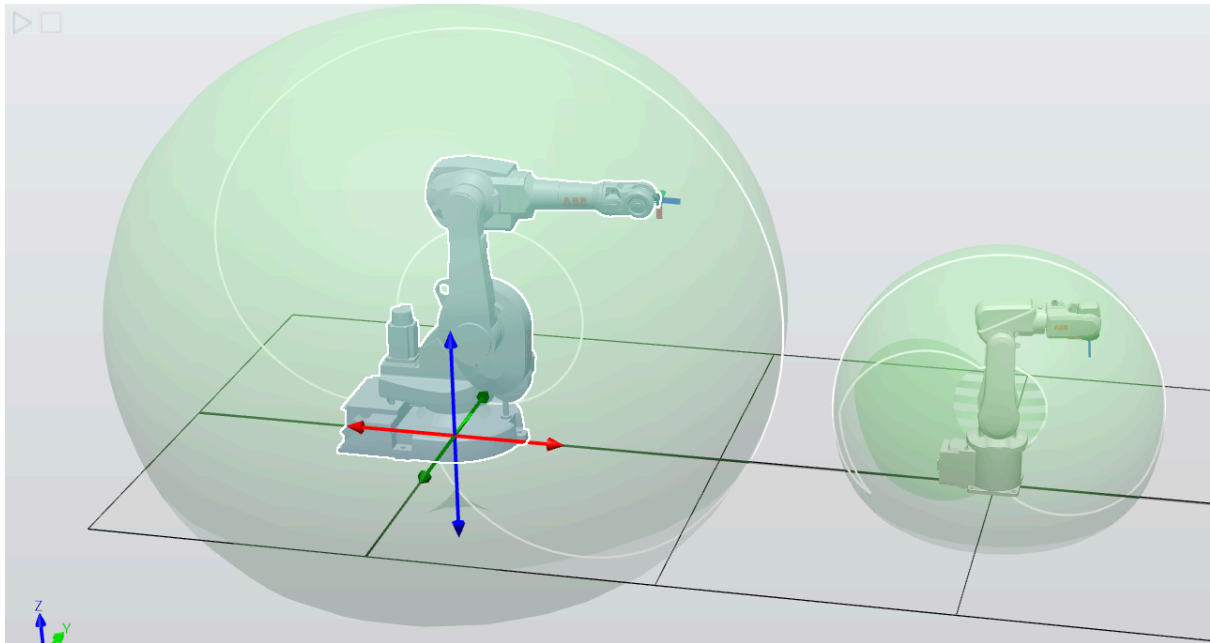
Question 1: Using the jogging function and the mechanic model of the robot, draw the kinematic model of IRB1600 with joints, links, and the type of the robot configuration.



The kinematic model for IRB1600 will remain the same as IRB120 as all the links are joints are similar for both of them.

Robot configuration for IRB1600 is also same for both of them which is Jointed arm.

Question 2: Enable the working envelope of the IRB1600 and try to move the End Effect of the robot outside the envelope. Are you able to do it? Explain your answer below:



Both robots have different size of working envelope but as function of an envelope is to restrict robot from going outside so that it does not create any interference with robots like joints and links or safety purposes.

Question 3: Compare the envelope of IRB1600 and IRB120. Can IRB120 do the task of IRB1600 and reverse? Explain your answer

In terms of Envelope size IRB1600 is a clear winner with larger area of 1.2 meter it can cover. Other specification like weight handling capacity will also increase with this. We'll need more area to setup this bigger robot.

In term of task, both robots can perform same tasks if it's within the capacity of IRB120 as IRB120 is smaller and covers less area for work and weight.

Week 3

MTRN 3060

PRACTICAL WEEK 3

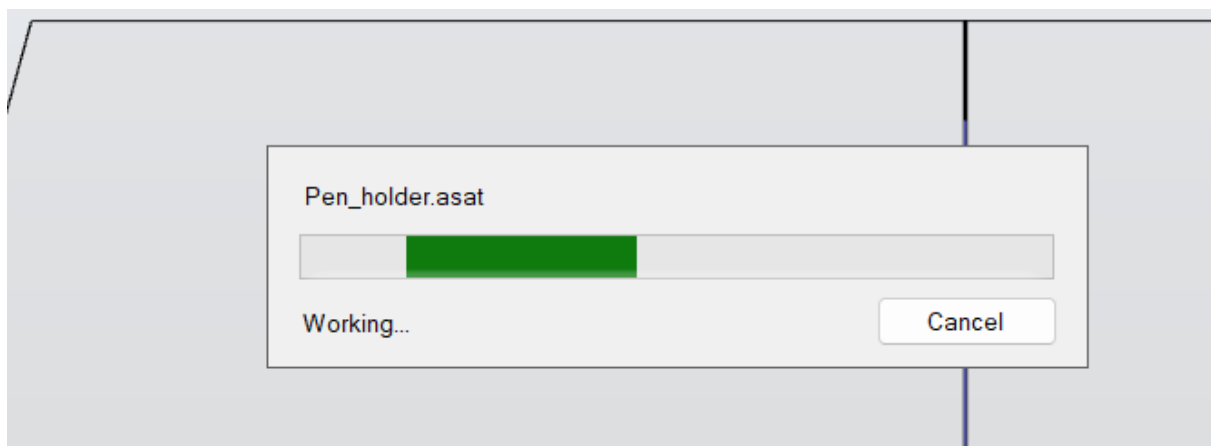
Introduction to Robot Tool and End Effect

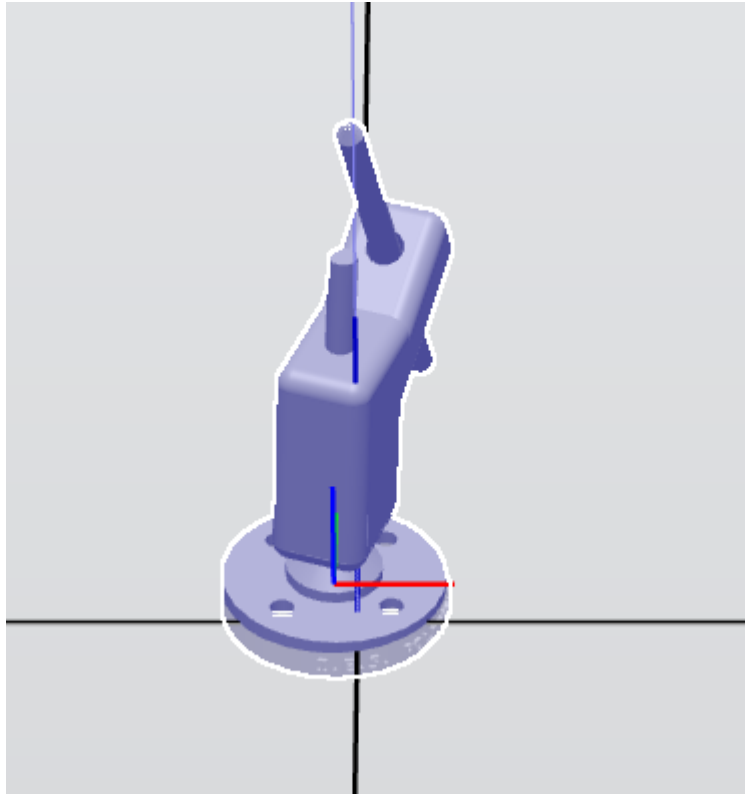
1 Build your First Robot Tool

1.1 Geometry Preparation and Create Tool

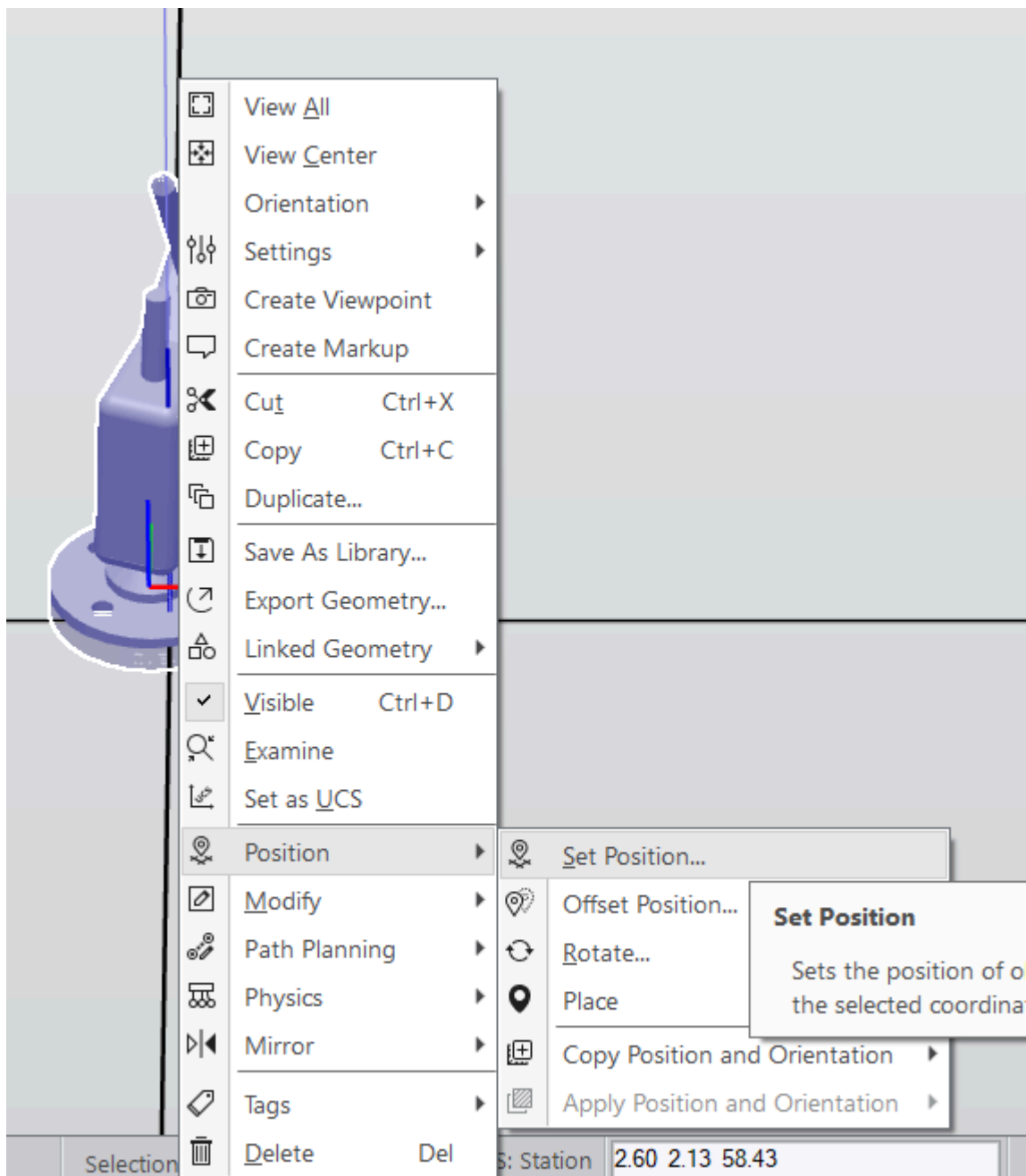
1.1.1. Create a new empty station

1.1.2. Drag the file name "pen holder.asat" to your Station to import the tool geometry





1.1.3. Move the imported CAD to the centre position by right click on it →
position → set
position → set all value to 0



Set Position: Pen_holder

Reference
World

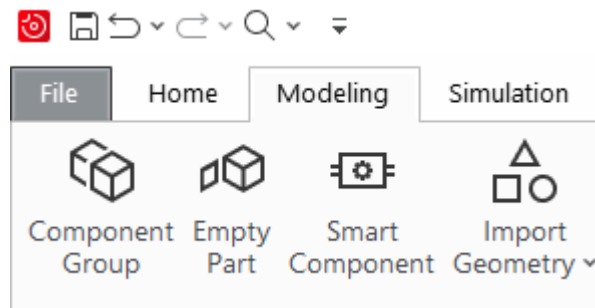
Position X,Y,Z (mm)
0 0.00 0.00

Orientation (deg)
0.00 0.00 0.00

Apply Close

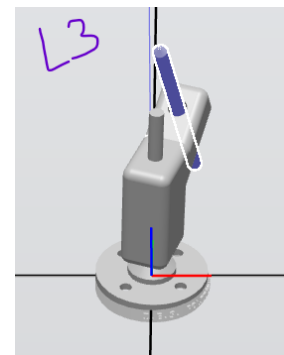
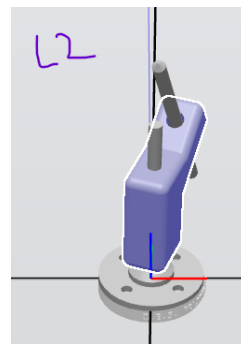
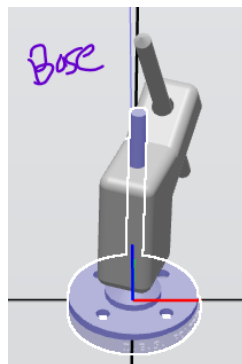
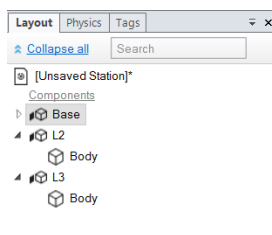
1.1.4. On the top panel → Modelling Tab

1.1.5. Create 2 empty parts



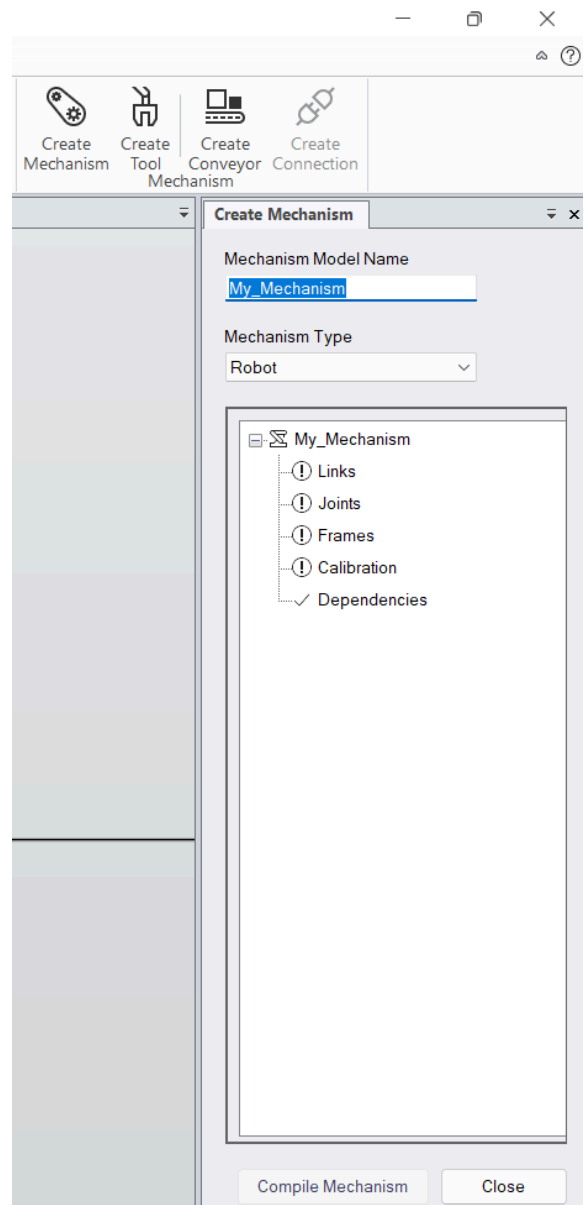
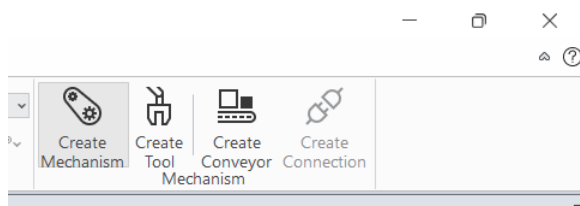
1.1.6. Drag and drop body in the pen holder part to empty part to separate it

1.1.7. Separate 3 tool bodies to each part and rename corresponding to the position of the body

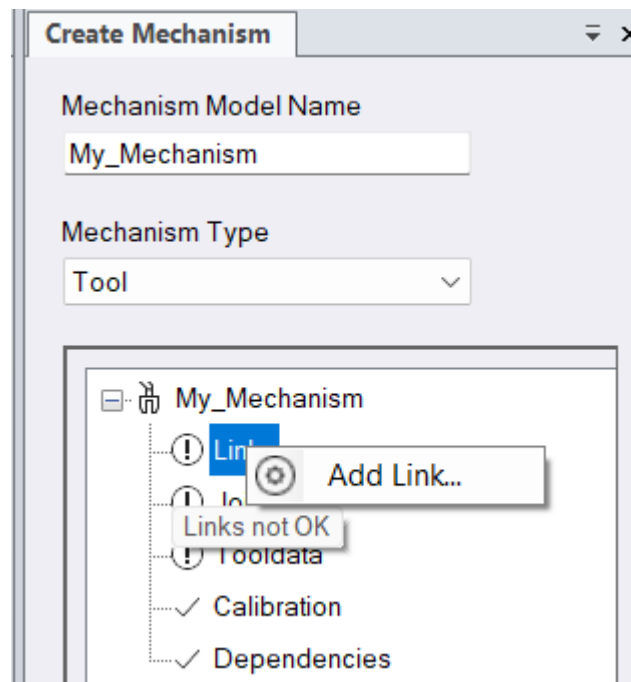


1.1.8. This pen holder has few links and fixed joints, so you will need to create a mechanism

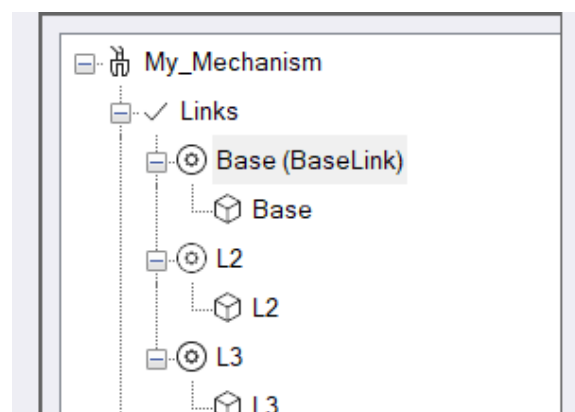
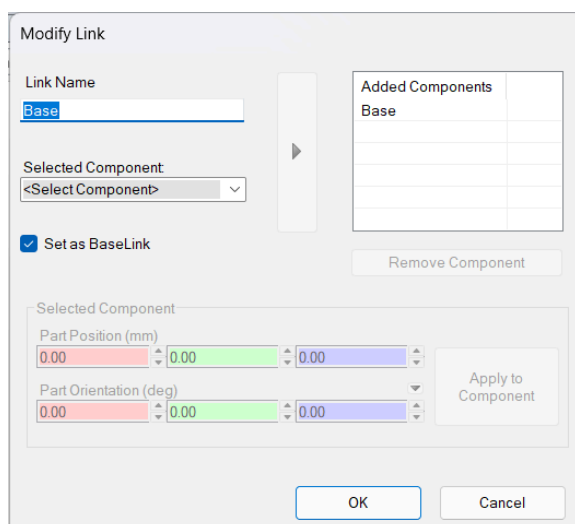
according to the real model: Highlight the imported geometry → modelling tab (top panel) → Create Mechanism-



1.1.9. Mechanism Type is "Tool" → right click → add link



1.1.10. Create 3 links: Link name should correspond to the position of it → Select one link at a time → green arrow → Tick "set as baseline" if that link is the base one → Apply



Modify Link

Link Name

Selected Component
 <Select Component>

☐ Set as BaseLink

Remove Component

Added Components

L2	

Selected Component

Part Position (mm)

Part Orientation (deg)

Apply to Component

OK Cancel

Modify Link

Link Name

Selected Component
 <Select Component>

☐ Set as BaseLink

Remove Component

Added Components

L3	

Selected Component

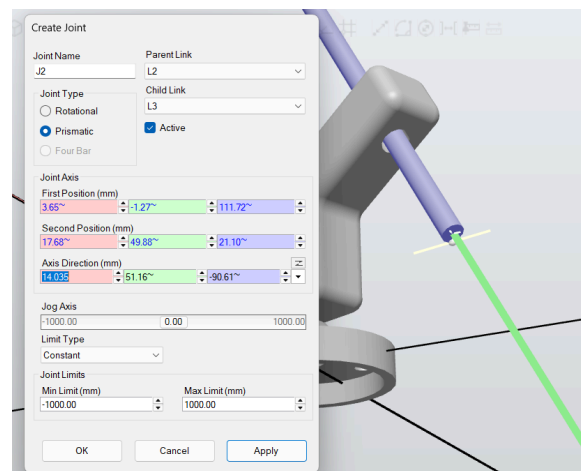
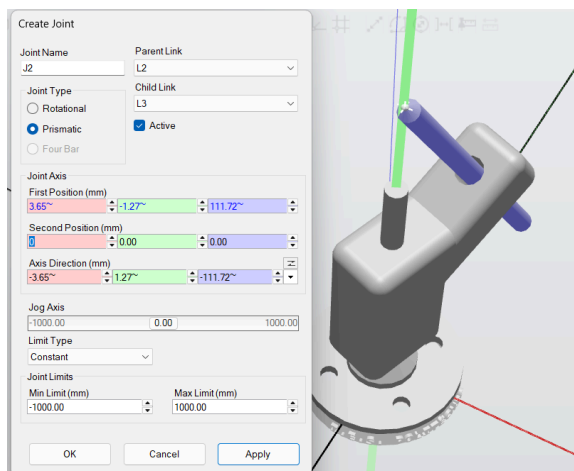
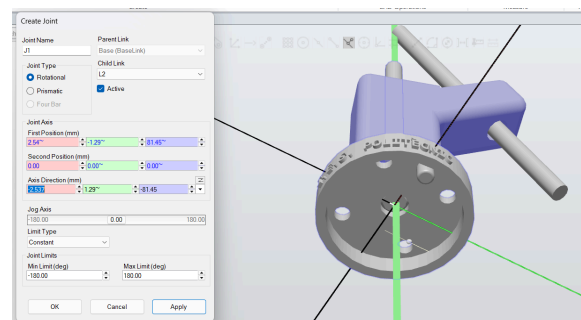
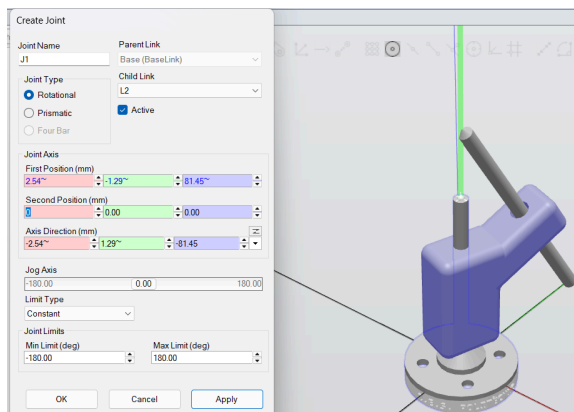
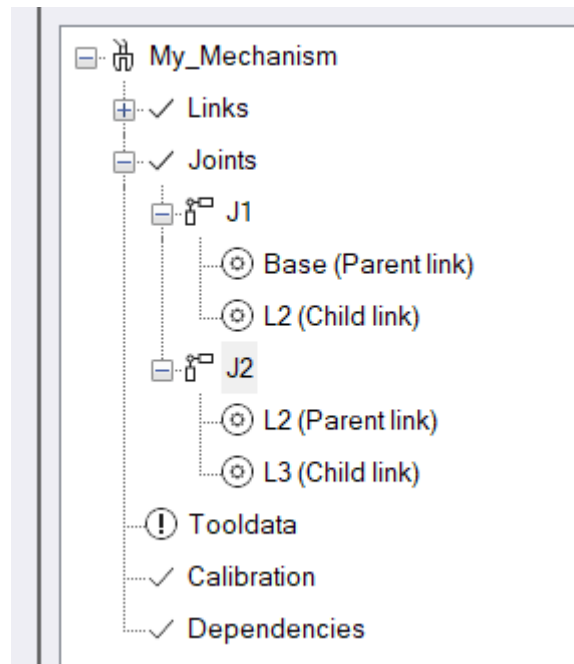
Part Position (mm)

Part Orientation (deg)

Apply to Component

OK Cancel

1.1.11. Create 2 joints between 3 links by double-click on Joints → Based on the pair of Link you want to have a joint between it, select ParentLink (act like a foundation for the Child link) and Child Link (position depend on the motion of the joint and foundation) → Select JointType base on how it should move → Select Joint Axis based on which direction it should move → Jog Axis to test the motion and make sure it looks similar to what it should do → Apply



Noted: highly recommend to use 2 points in joint axis to define the motion of the joint for better accuracy

1.1.13. Tooldata: Put 1 into the centre of gravity X section → press OK- This will create an end effect frame for the tool but at the wrong position- following the next section to correct this.

Create Tooldata

Tooldata name:
My_Mechanism_1

Belongs to Link:
L3

Position (mm)
0.00 0.00 0.00

Orientation (deg)
0.00 0.00 0.00

☐ Select values from Target/Frame
<Select Frame>

Tooldata

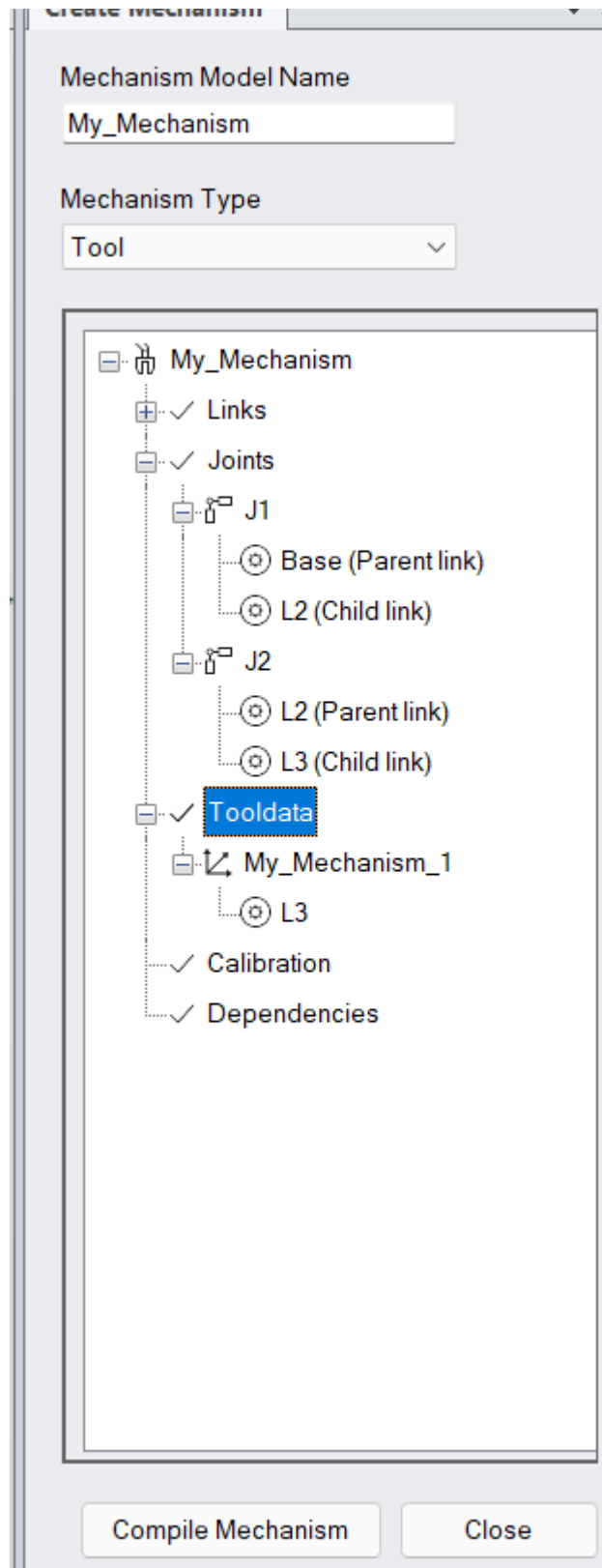
Mass (kg)
1.00

Center of Gravity (mm)
1.00 0.00 0.00

Moment of Inertia Ix, Iy, Iz (kgm²)
0.00 0.00 0.00

OK Cancel

1.1.14. Once you have double-checked every step, then proceed with caution (can be irreversible):
press compile to finish designing the tool



2 Reference Frame and Tool

In this section, you will practice with fixed joints that are created in the previous section and mounting robot tools to the robot flange.

Usually, the first stage is designing the tool in the simulation and then producing the end effect frame from it. However, the end effect frame in the simulation might be misaligned due to multiple factors when the tool is mounted on the robot, so FlexPendant is used to produce another end effect frame.

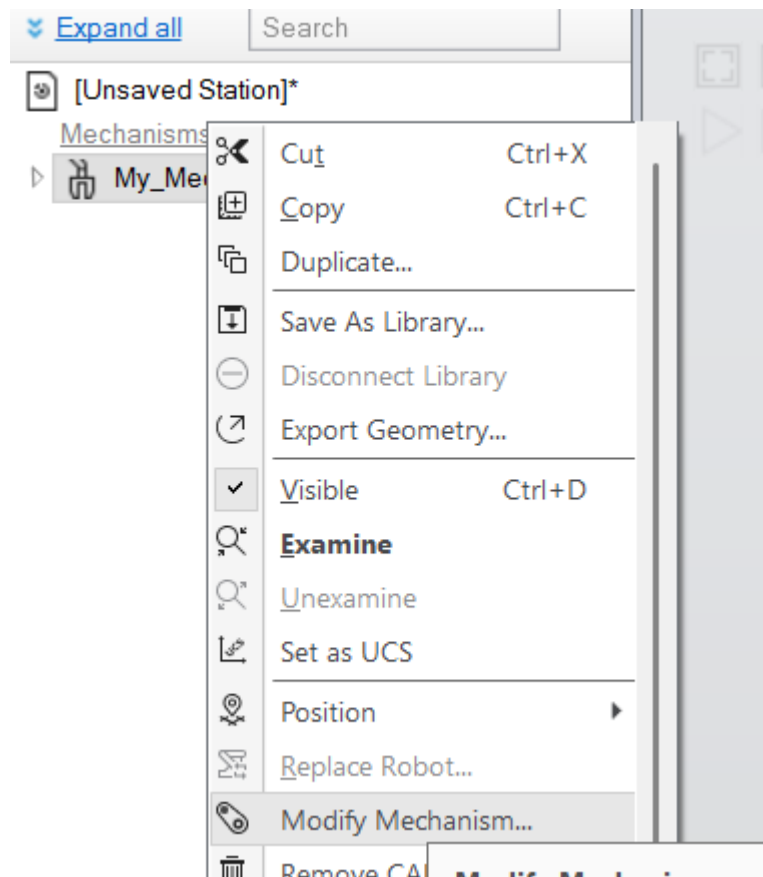
2.1 Direct Tool Positional Data Entry- Using the simulation

The following step shows how to modify the tool frame in the simulation. The tool frame defines the position and orientation of the robot's end-effector (the tool).

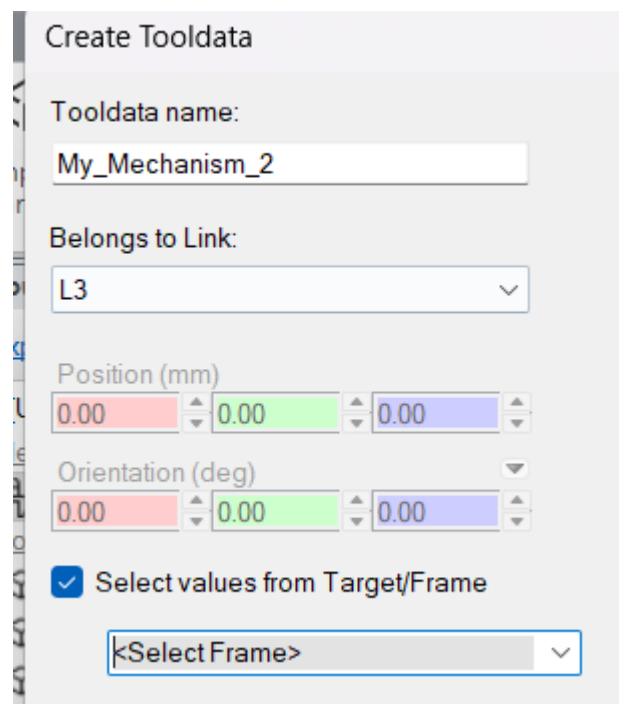
2.1.1. The end effect frame of this tool should be placed at the end of the pen tip.

2.1.2. To fix this: layout tab (left panel) → right click pen holder tool → Modify Mechanism

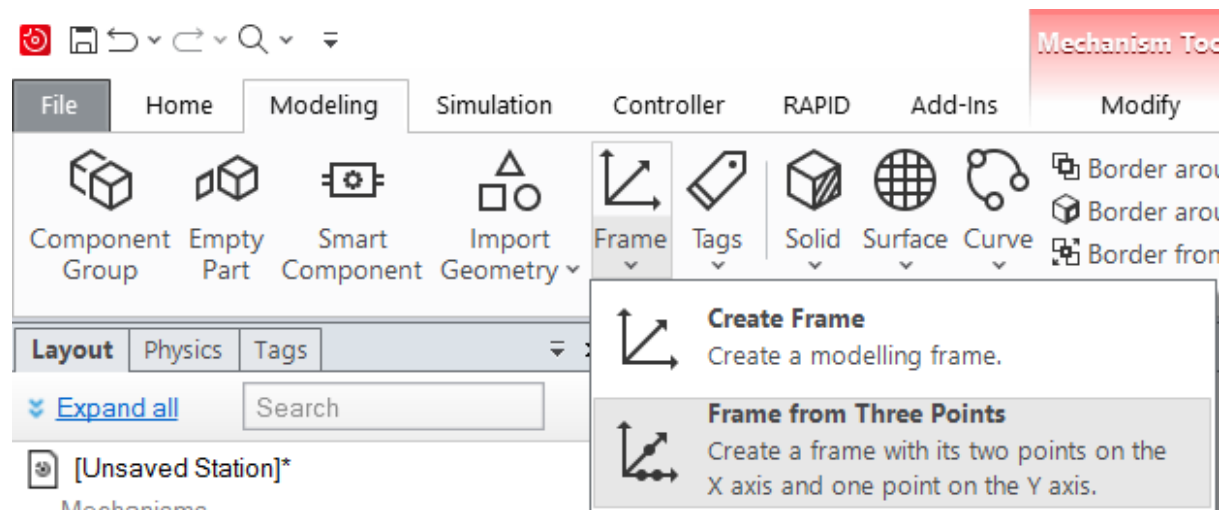
→A panel pop up on the right side



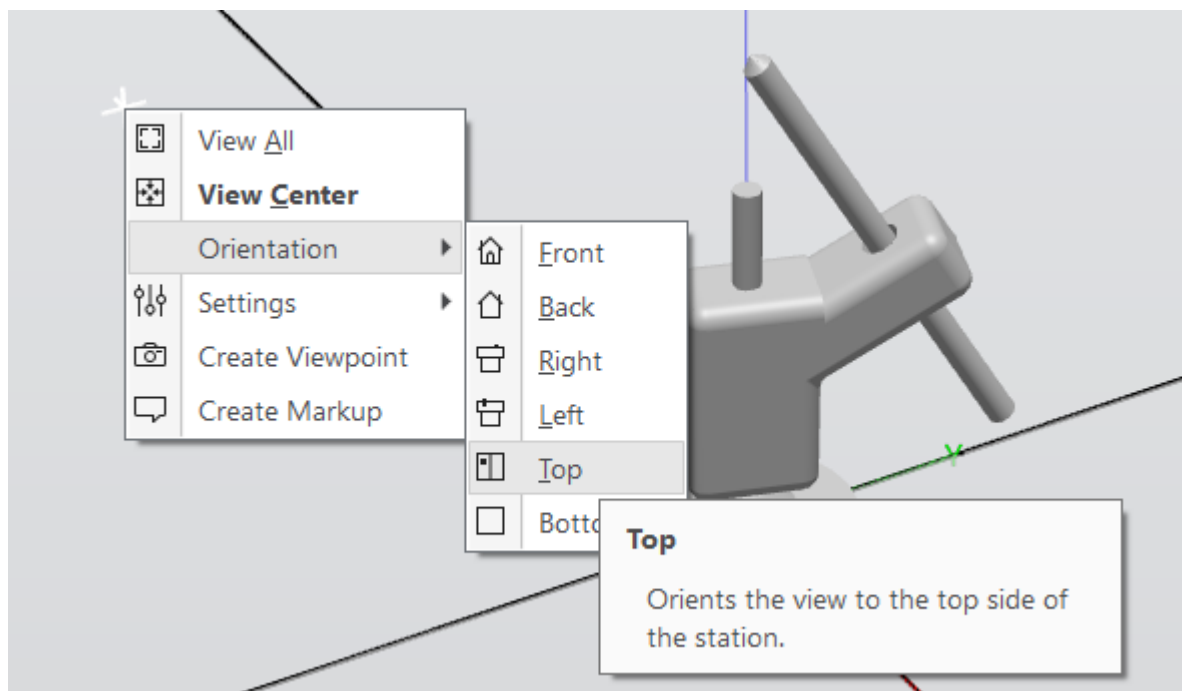
2.1.3. double-click on the Tooldata to create another end effect frame → tick on the box to use the existing Frame → there is no frame at the moment

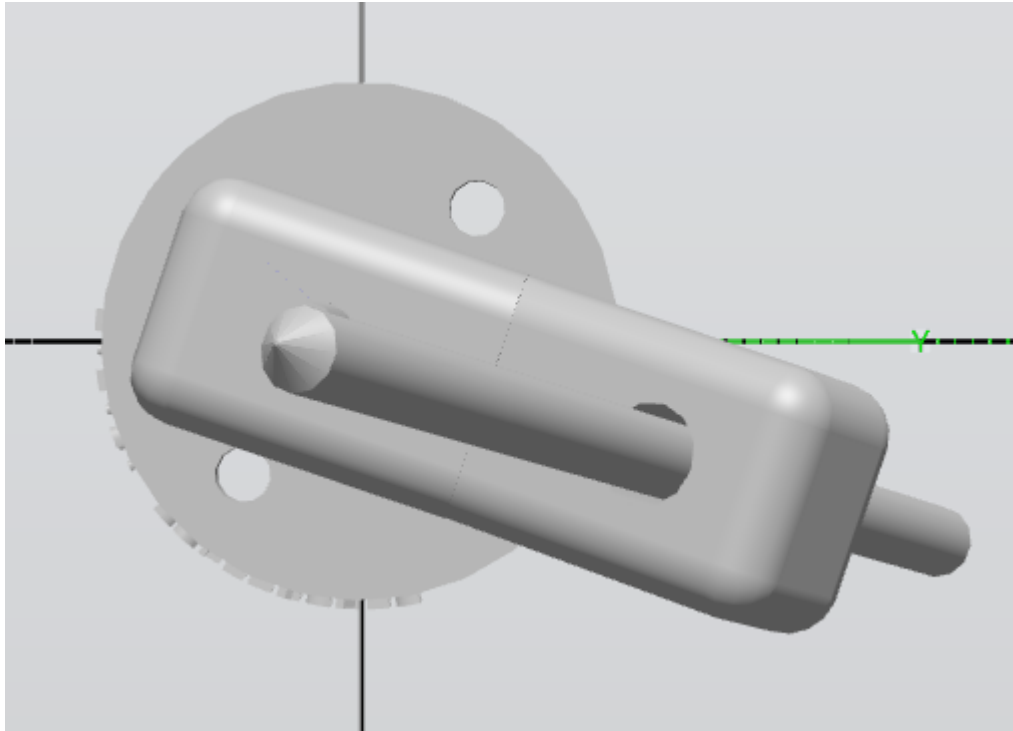


2.1.4. To create a reference frame for the Tooldata → Modelling tab (top panel)
 → The little
 arrow below "Frame" → Frame from Three Points



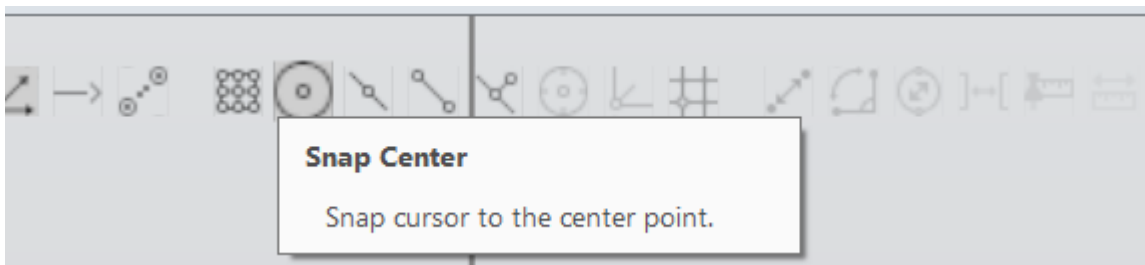
2.1.5. inside the simulation view, → right click → orientation → Top- To go to top view

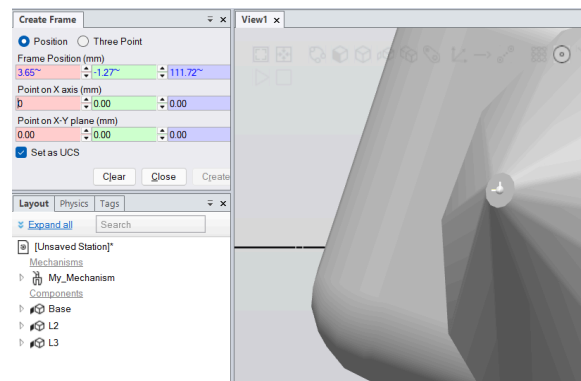
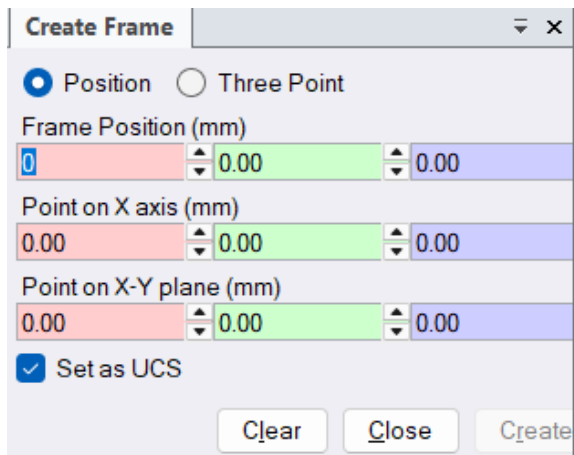




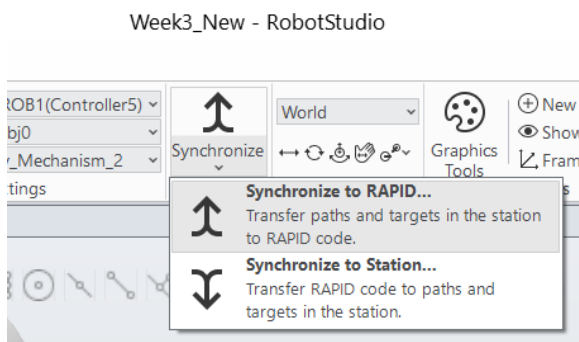
2.1.6. Use the snap centre selection mode- to be able to select the centre of the tip

→ Click on the red box (X axis) below Frame position → Click in the centre of the pen tip (you will see a small white sphere)



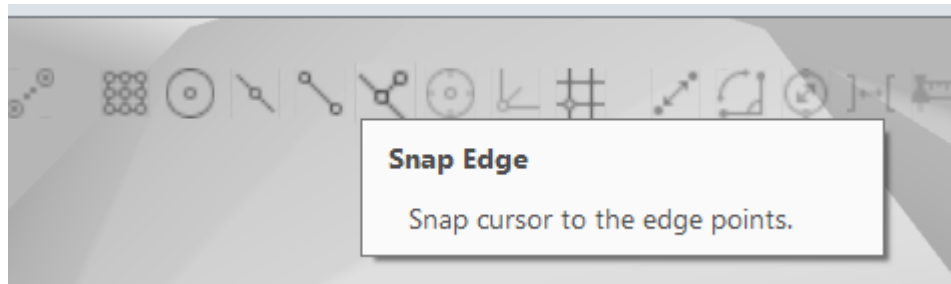


- go to the home Tab → click small arrow below synchronize → synchronize to RAPID

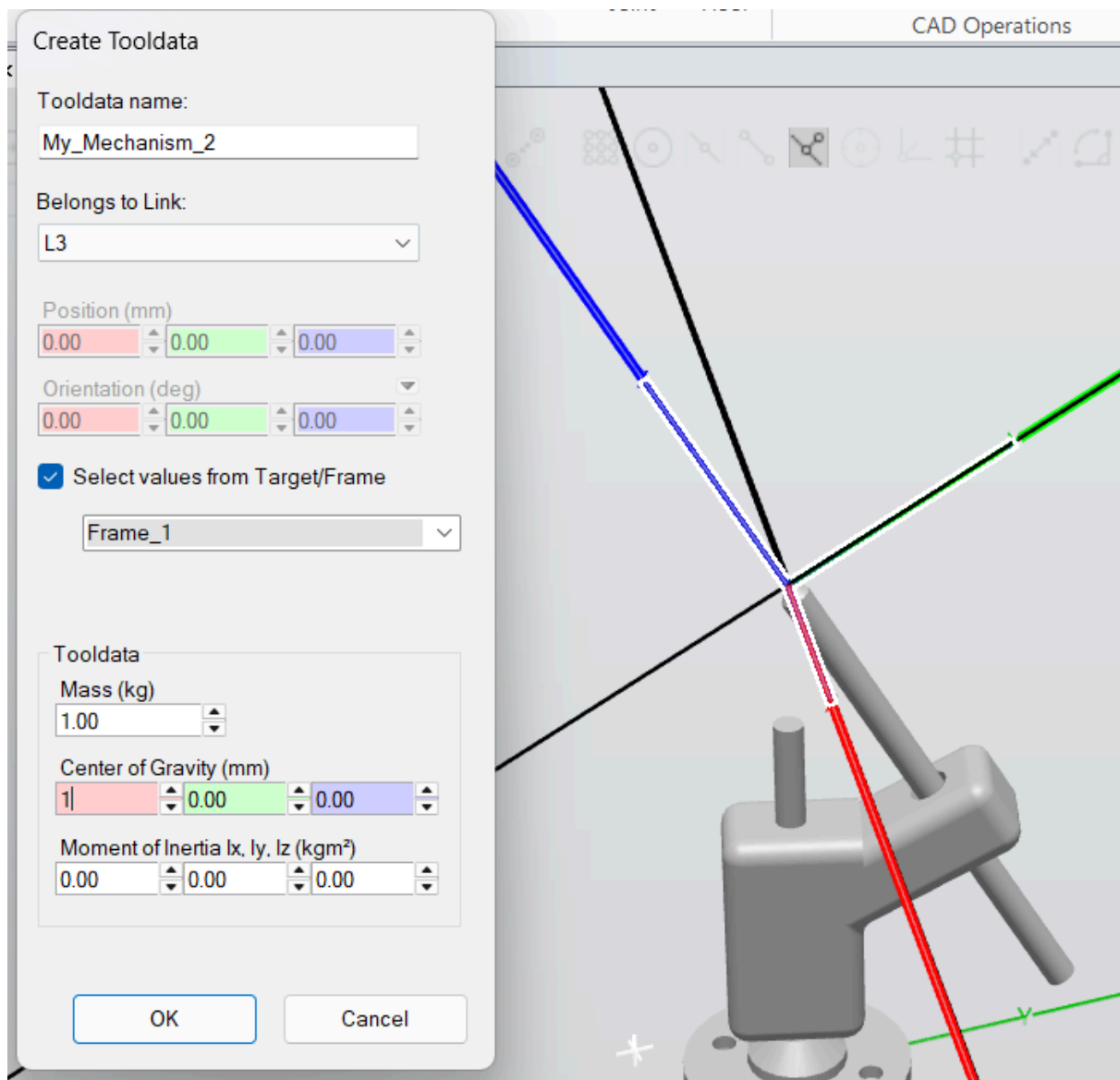


2.1.7. Switch to snap edge mode to select the edge of the pen tip for the Second and Third points of the Frame

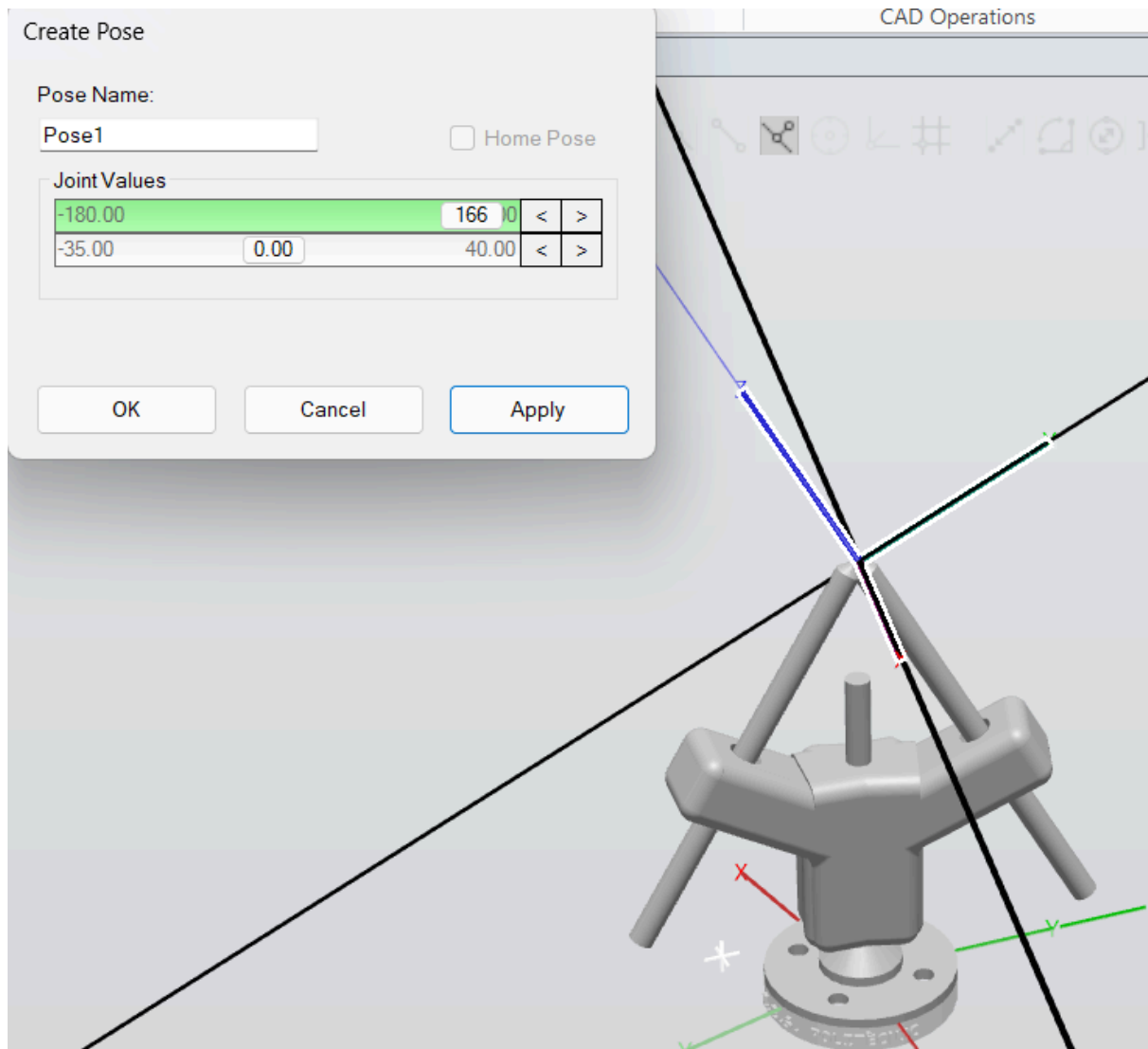
2.1.8. Press create → a new XYZ frame named "Frame 1" is placed on top of the top, but it might have the wrong orientation → Right-click on the frame inside the layout tab → Rotate → Try to rotate the frame so the Z-axis pointing outside the pen.



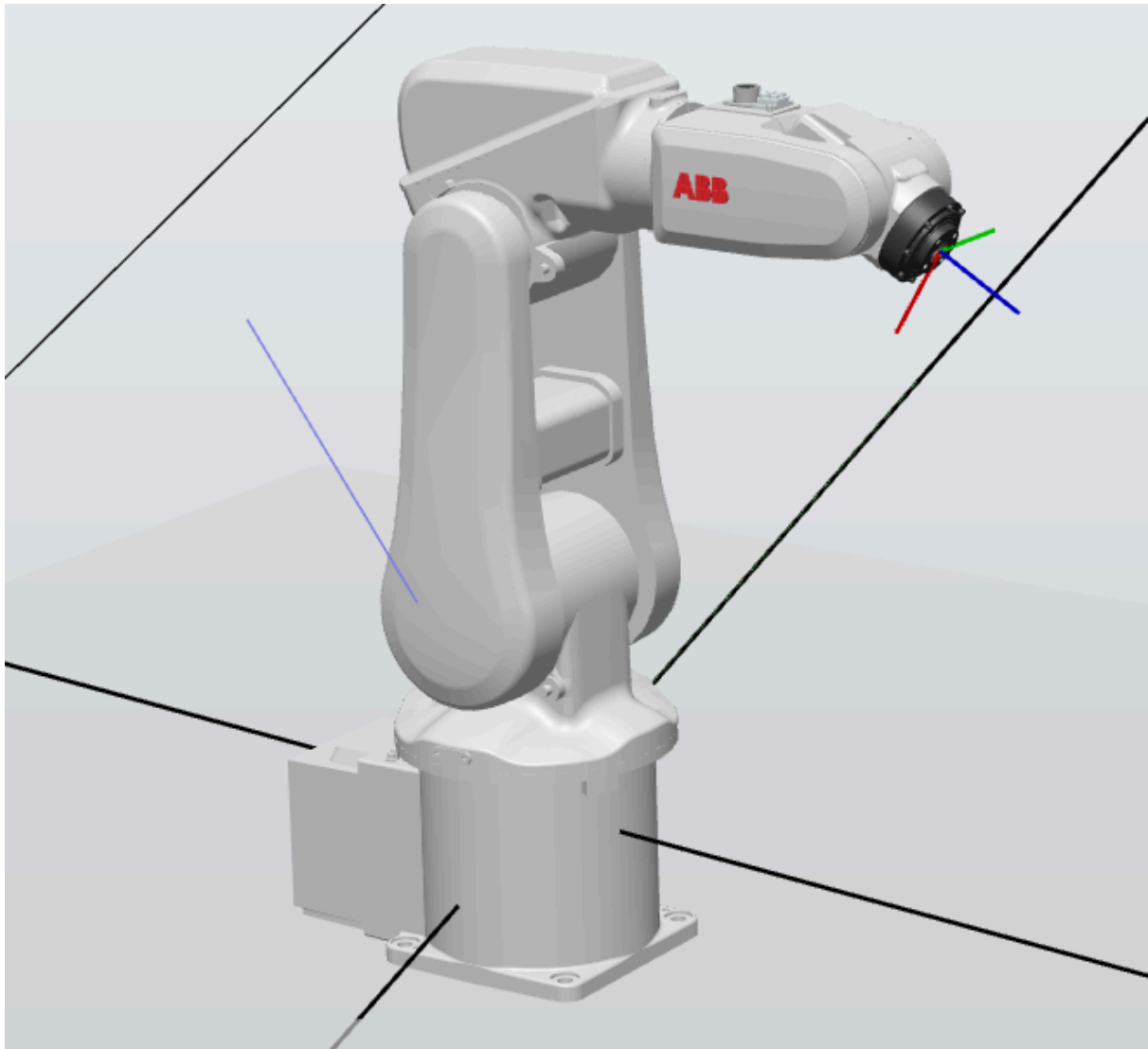
2.1.9. Back to the Tooldata windows → select "Frame 1" → OK → a new tooldata for end effect frame will be created



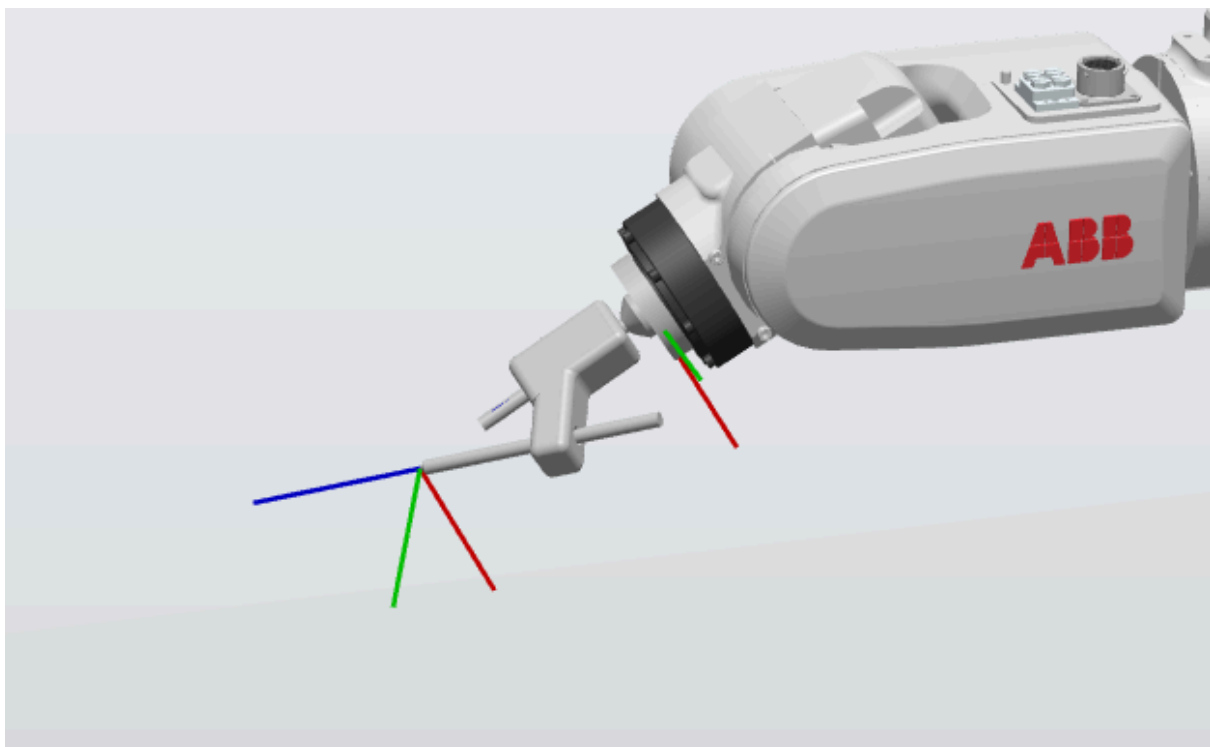
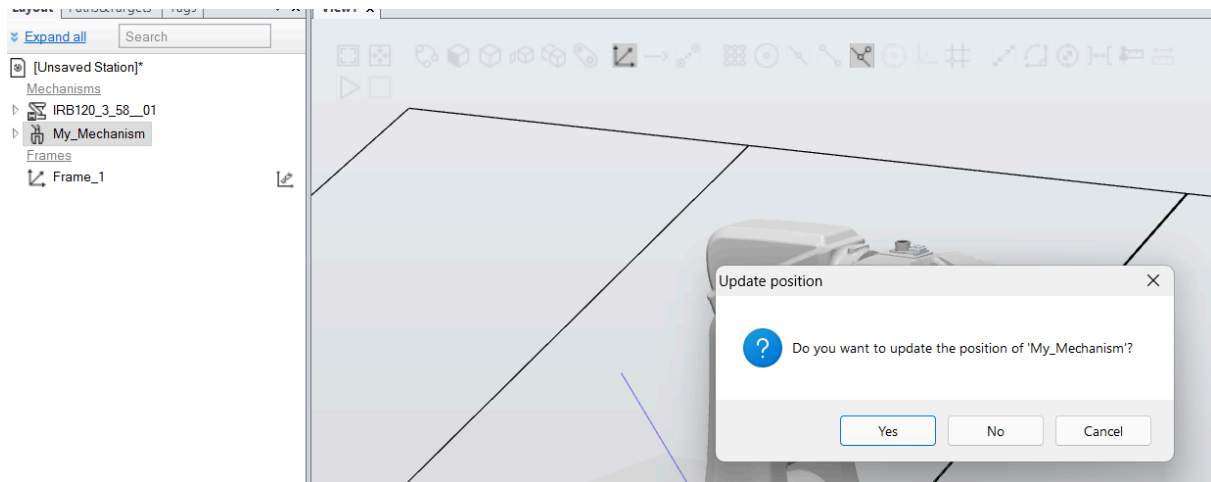
2.1.10. Add a few poses to the tool, so the end effect can move with the tool motion when the shape is changed. → press compile



2.1.11. Add an IRB120 with its controller to the simulation

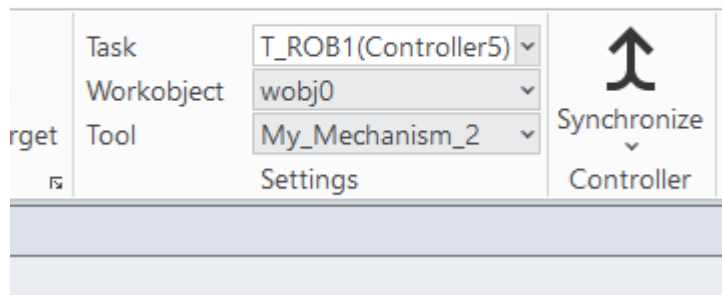


2.1.12. In the left panel → Layout tab → Mount the tool on the robot by dragging it to the robot, → press yes to update the position → The tool is mounted in front of robot flange

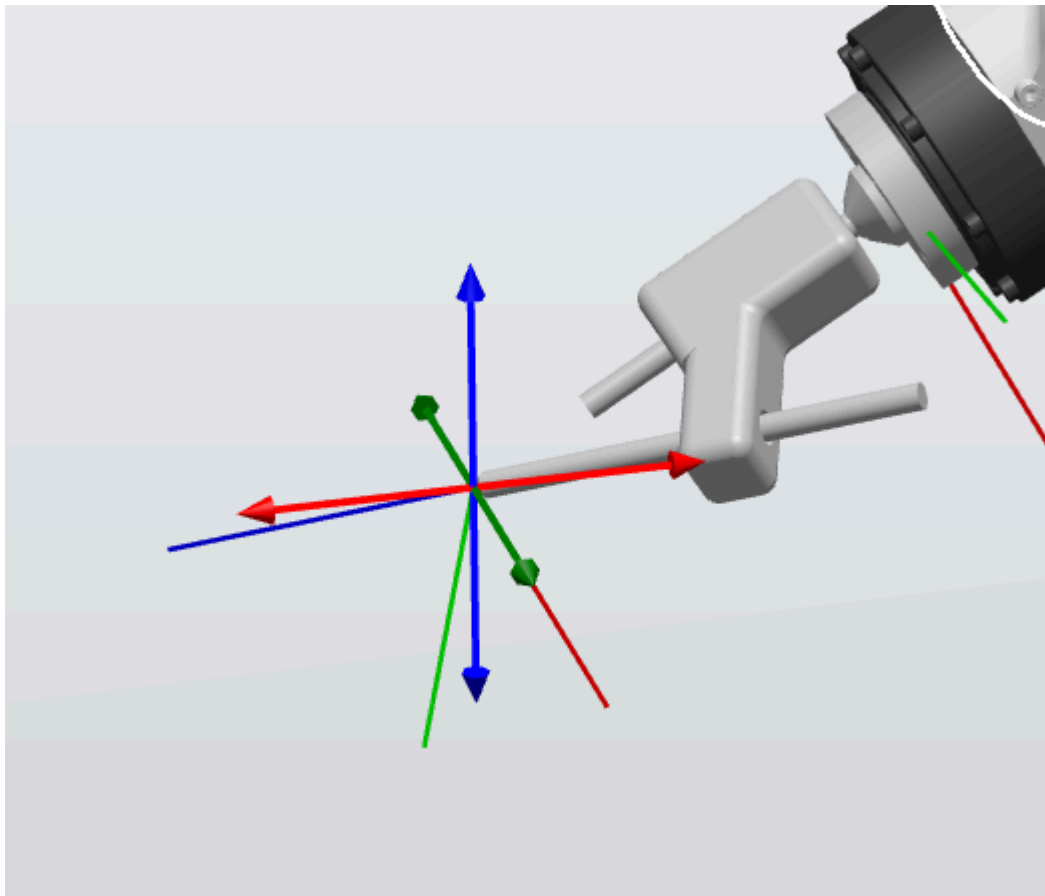


2.1.13. Because you have two tooldata, you need to select the correct one, so the simulation can run properly. To switch between tooldata: Home tab (top panel) → Tool → little arrow → select the tooldata.

[Unsaved Station] -



2.1.14. To verify your end effect position → click on linear jog mode → click on the robot → if you see the drag arrow on the tip of the pen holder then it is correct.



2.1.15. Noted: The location and orientation of this frame will not match with the pen tip on the real robot in the lab due to many factors.

2.2 Teaching Tool Centre Point (TCP) using the FlexPendant in real world

- TCP teaching/calibration is done manually by moving the robot
- letting the TCP tip against a fixed point in the environment.
- The fixed point is typically a small point on the teaching tool, like the tip of a nail
- Default TCP Orientation (another name is 3-point method): use the same rotation as the original centre point but allows the programmer to define a new location for that centre point.
- This method works with tool that has no angular adjustments like grippers or vacuum systems that aligned parallel to the mounting face

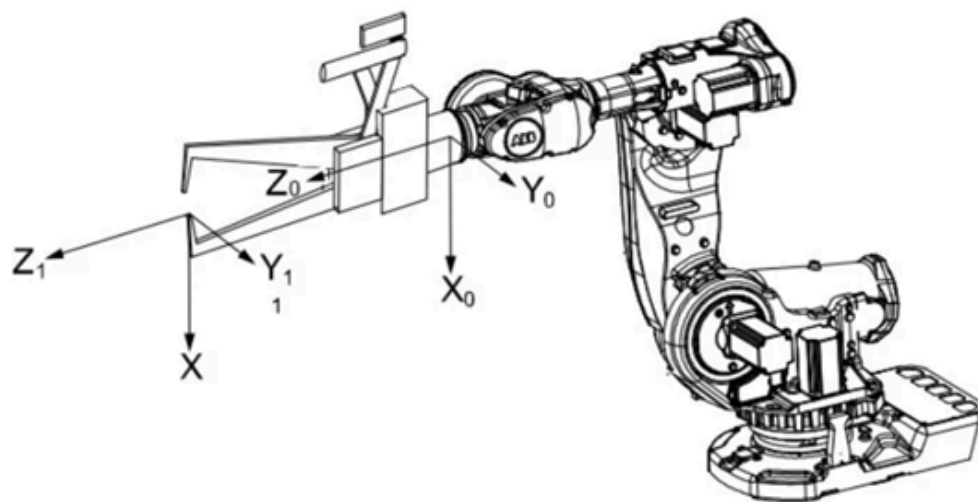
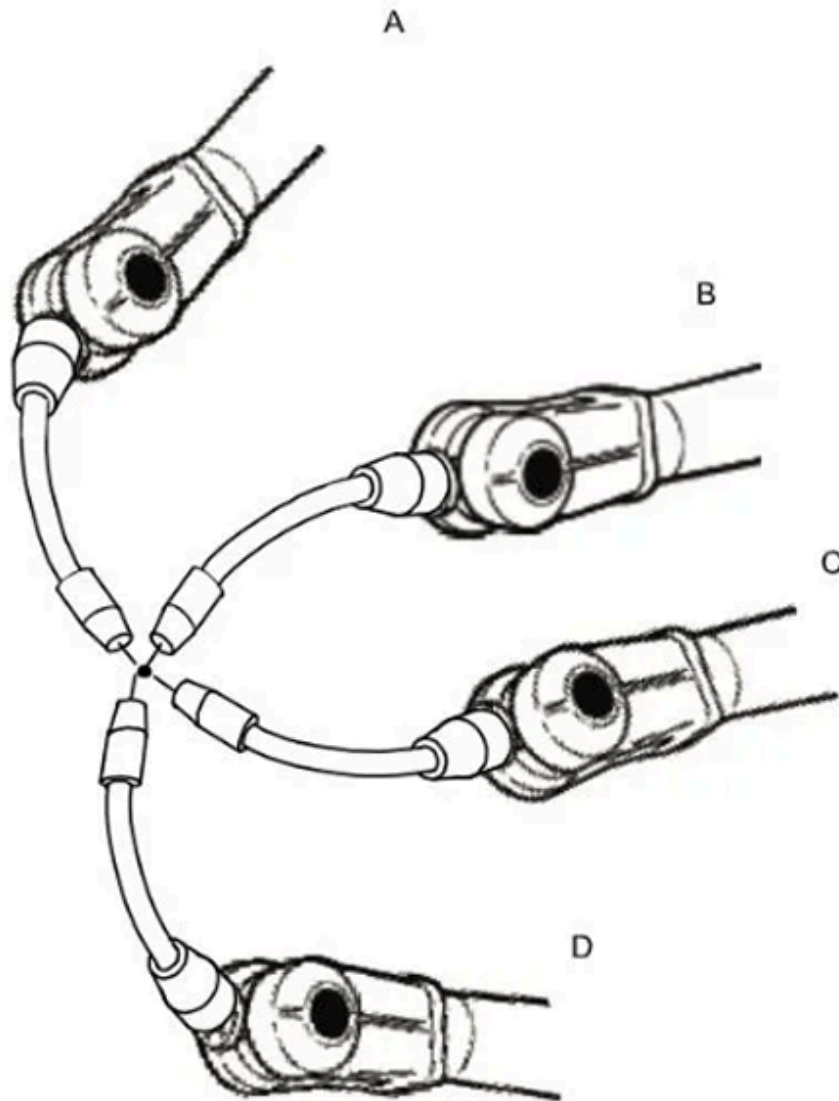


Figure 2.5: Tool with no angular adjustment

- TCP&Z: Additional of TCP method, which establishes orientation by teaching an additional position to define the z-axis.

- Final teach motion should jog the robot solely in the z-axis direction, away from the teaching tool.
- This option will set which way Z+ will be oriented, with the x-axis and y-axis to follow.
- This is the minimum requirement for the tool that has angular adjustments like the Pen holder

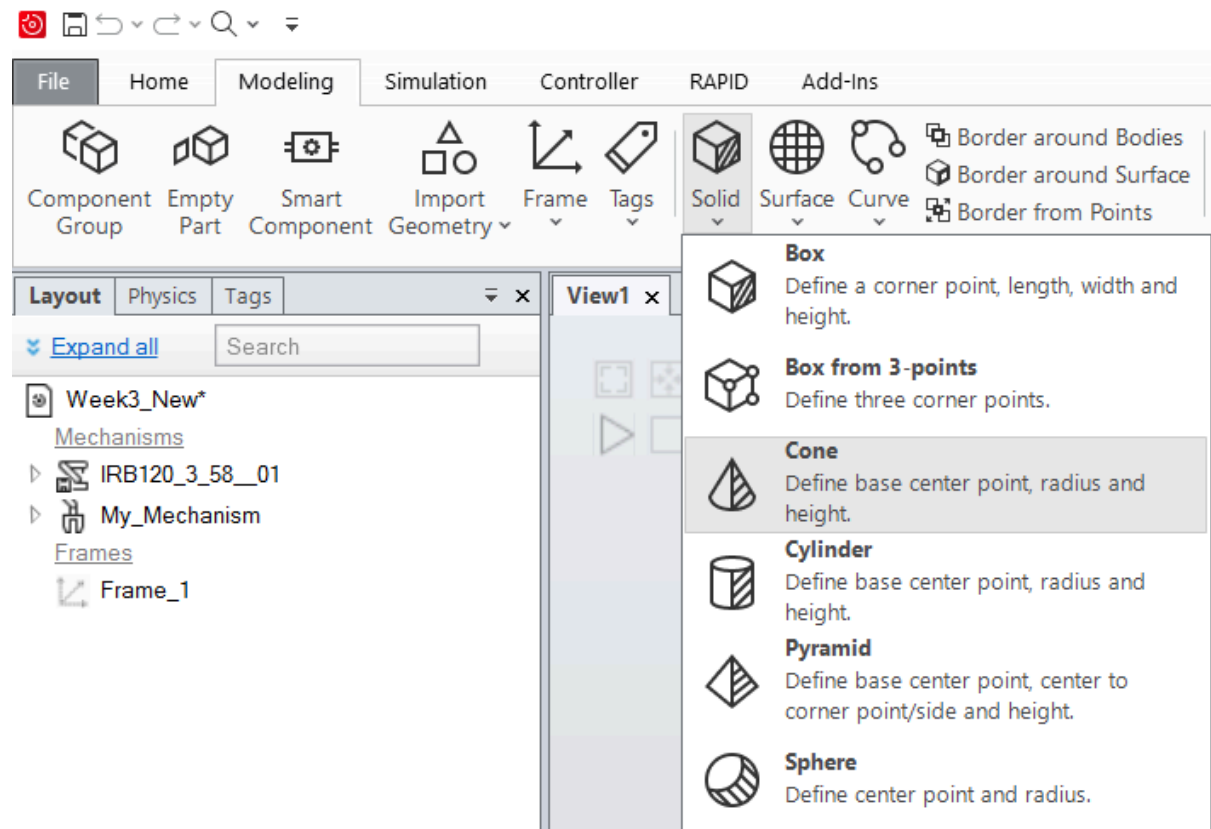


The following sections will show you the way to practice calibrating the tool center position (TCP) in the simulation so that we can do it with the real robot in your minor project

2.2.1 Define the teaching tool

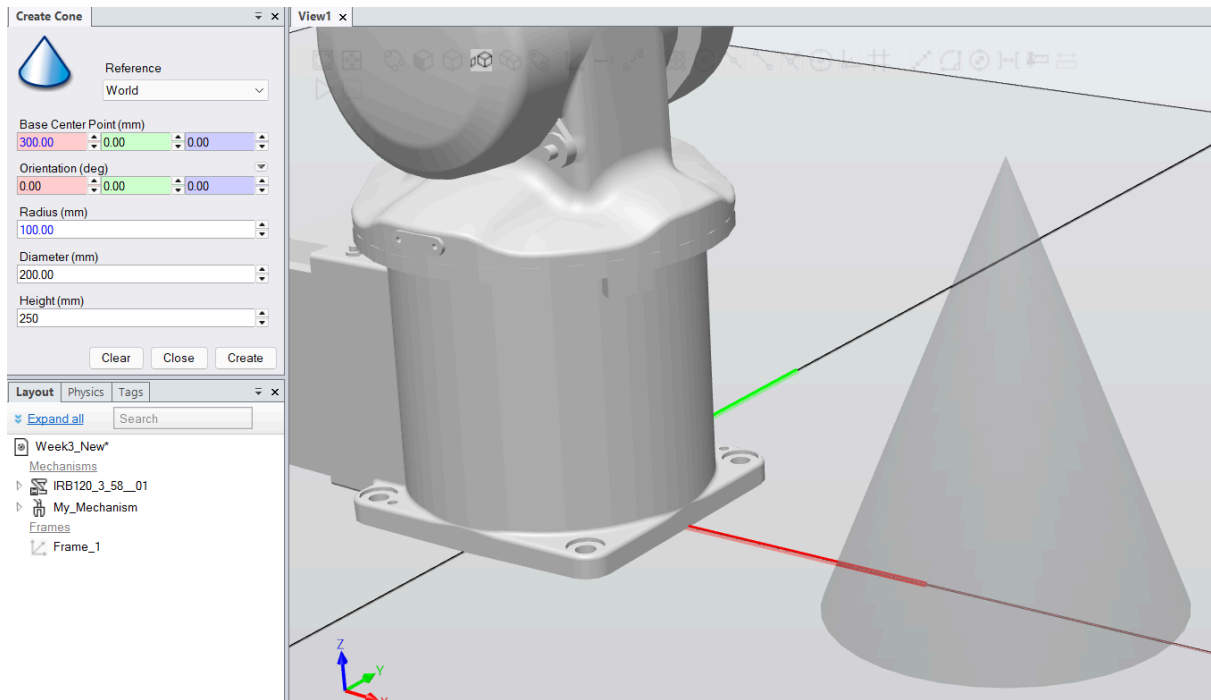
Before calibrating the TCP, you need to establish a reference point. We will create a cone and use its tip for this purpose.

2.2.1. Create a small cone with sharp tip by Modelling tab → Small arrow below Solid → Cone



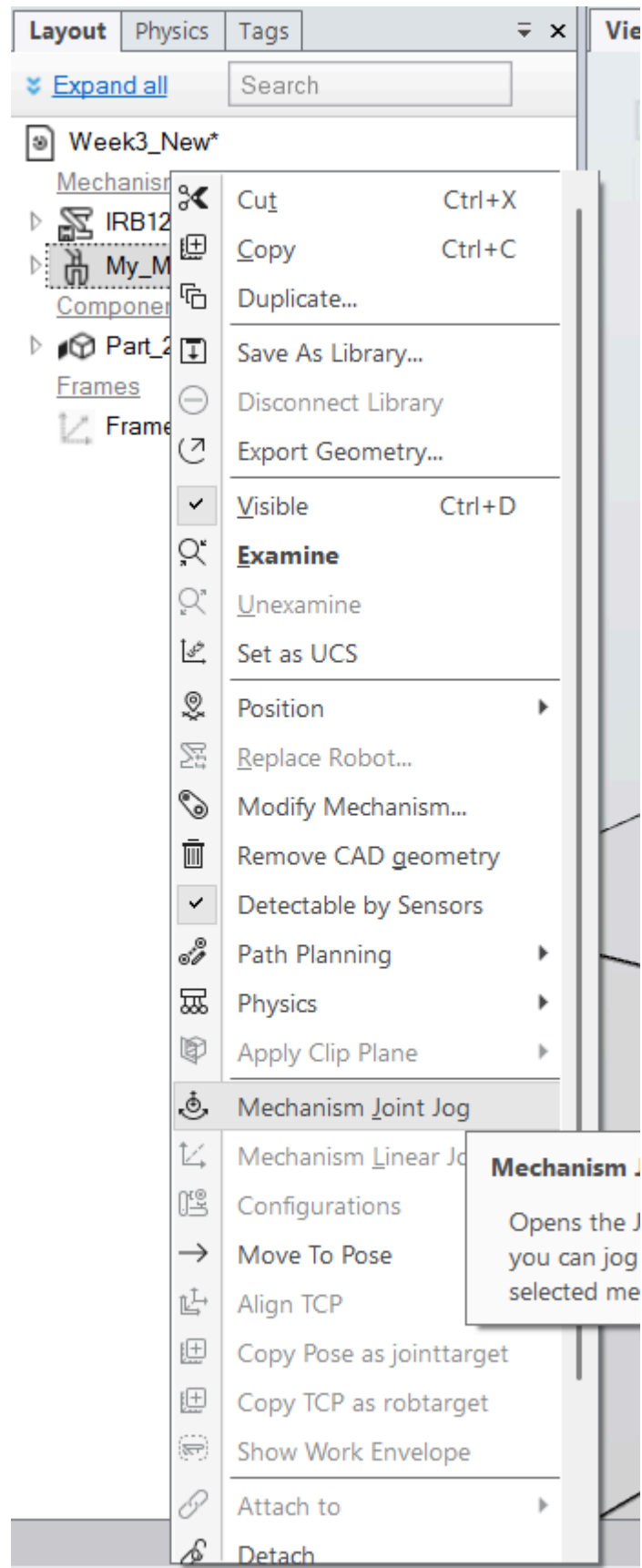
2.2.2. Set the cone base centre 300mm away from IRB120 robot

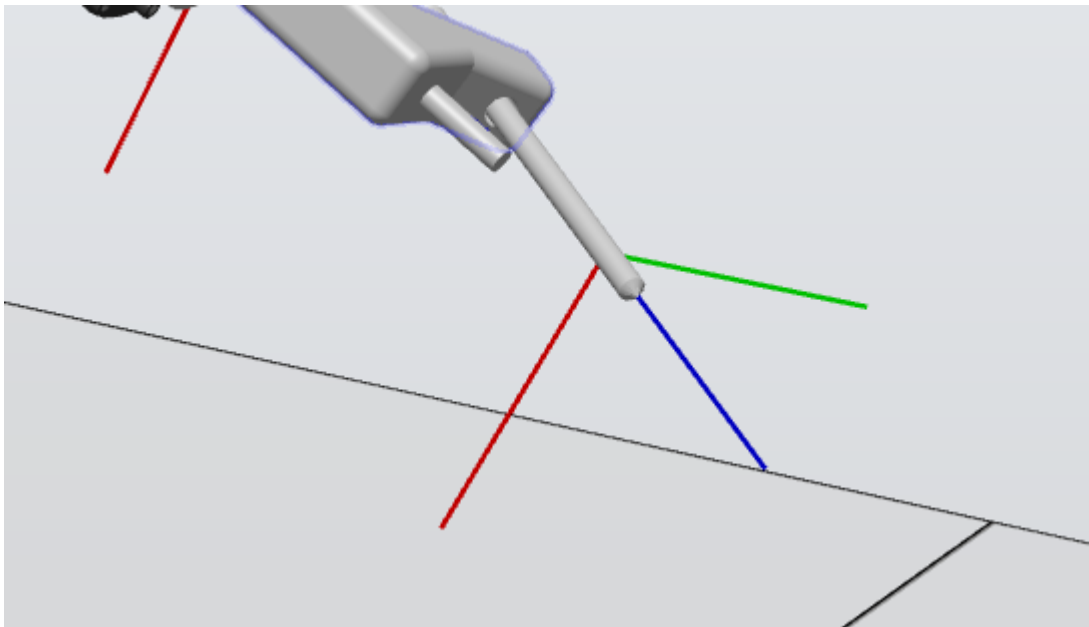
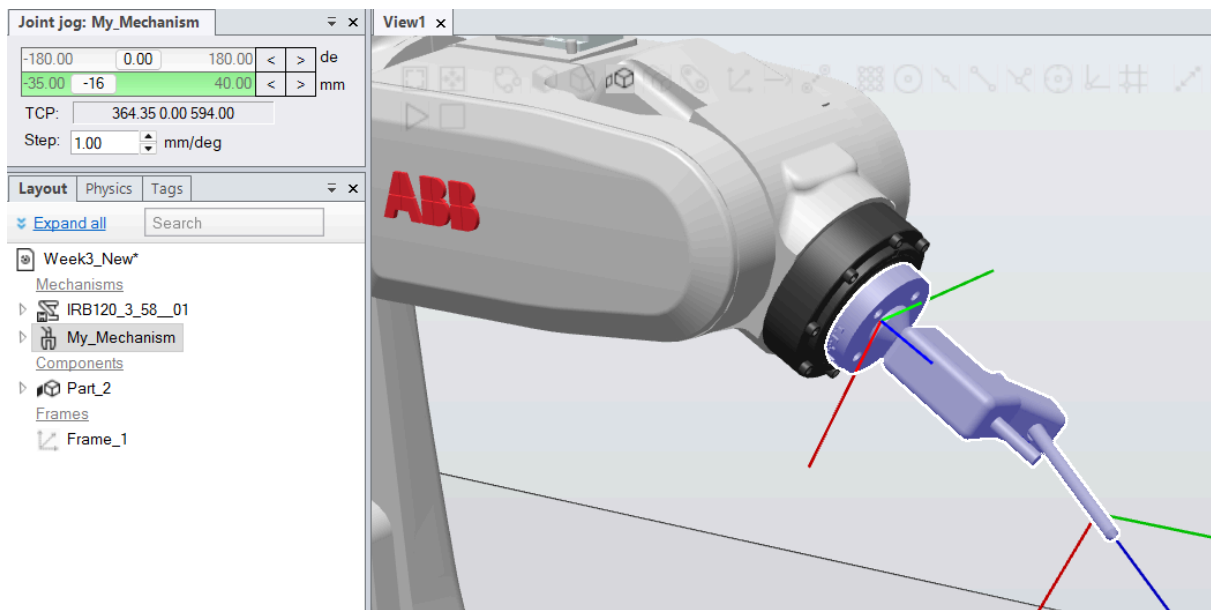
2.2.3. Select the value to create a similar cone in



2.2.4. To create a misplacement between the Pen and the Holder in simulation:
 Right-click on
 the Tool → Mechanism Joint Jog, → Push the Pen further out. Now, the end
 effect is in
 a different location compared to the TCP frame

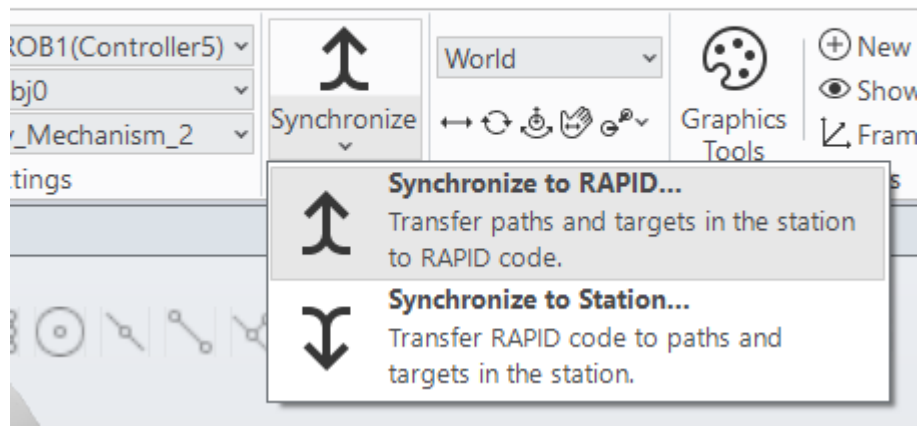
-



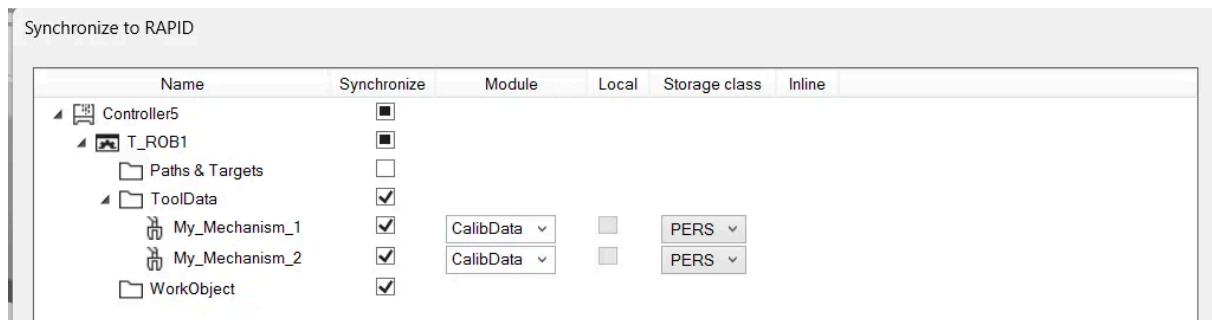


- Go to home tab → click on synchronize arrow → synchronize to RAPID

Week3_New - RobotStudio



- window will pop up you need to check the boxes as shown → Press OK

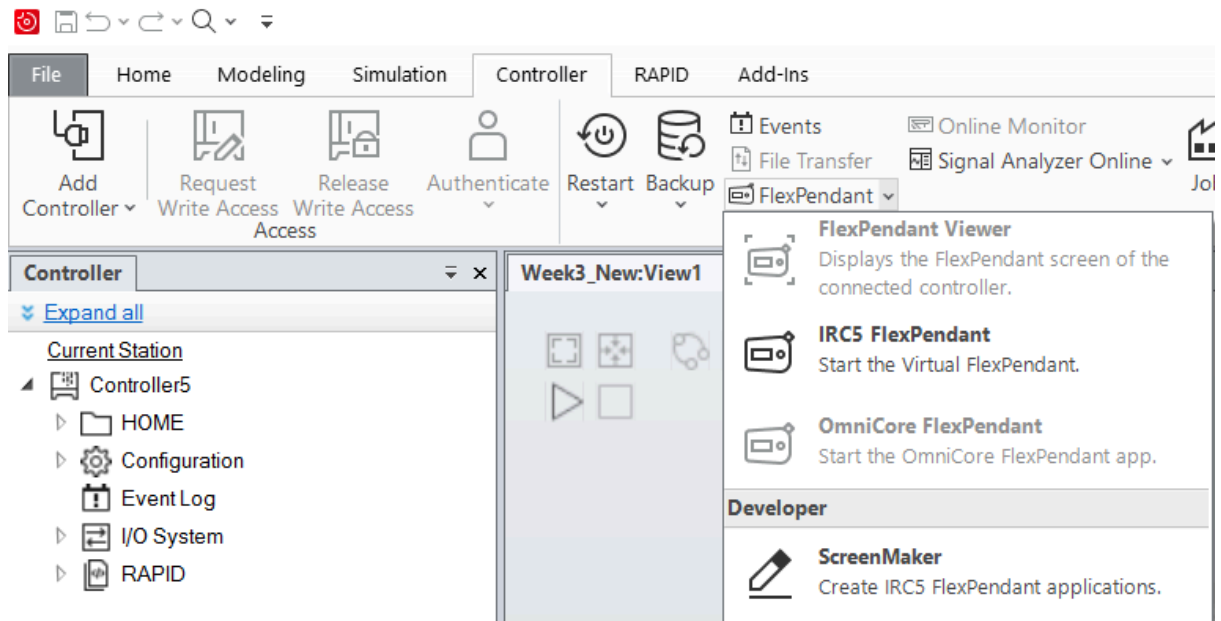


2.3 TCP&Z

The step below shows you how to calibrate TCP using TCPZ method, you will need to practice

a few times before you can produce the correct result:

- Go to controller → click on flex pendant → IRC5 FlexPendant



2.3.1. To access the Tooldata in the FlexPendant: Menu → Program Data → Tooldata

2.3.2. Follow the image to transfer TCP made from the last section to the controller. If every thing is done well, then it should look similar to

Auto
 Controller5 (KUNDAN-PC)

Motors Off
 Stopped (Speed 100%)

Data of type: tooldata

Active filter:

Select the data you want to edit.

Scope: RAPID/T_ROB1
 Change Scope

Name	Value	Module	1 to 3 of 3
My_Mechanism_1	[TRUE,[[0,0,0],[1,0...	CalibData	Global
My_Mechanism_2	[TRUE,[[3.62455,-1...	CalibData	Global
tool0	[TRUE,[[0,0,0],[1,0...	BASE	Global

New...
Edit
Refresh
View Data Types

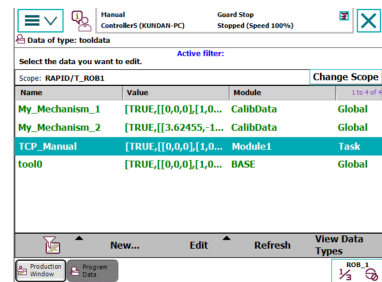
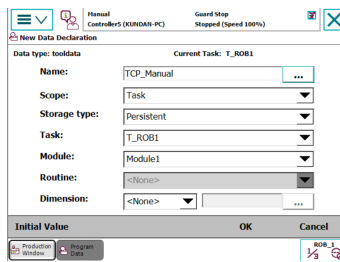
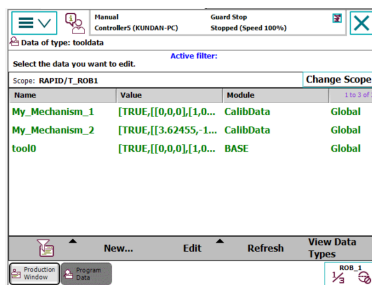
Production Window

Program Data

2.3.3. Switch FlexPendant from Auto to Manual Mode



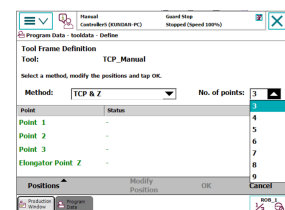
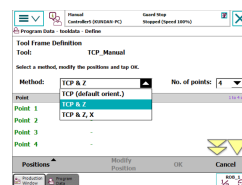
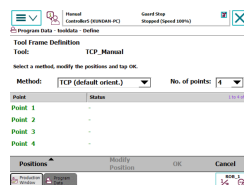
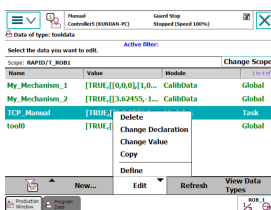
2.3.4. To create a new TCP: in the tooldata window on FlexPendant press New
 → Name:
 TCP manual → OK



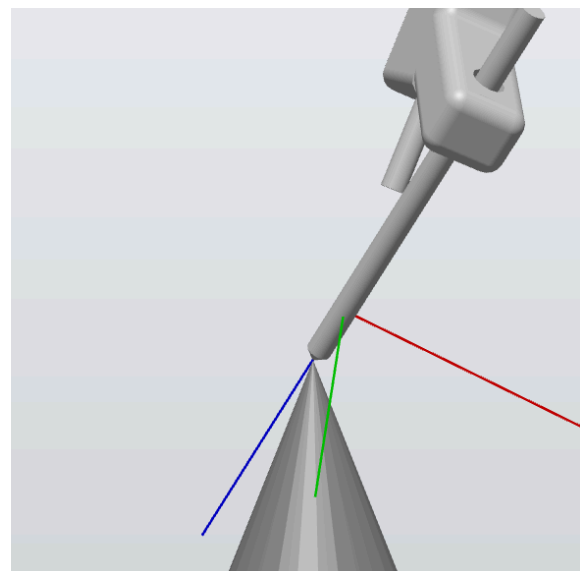
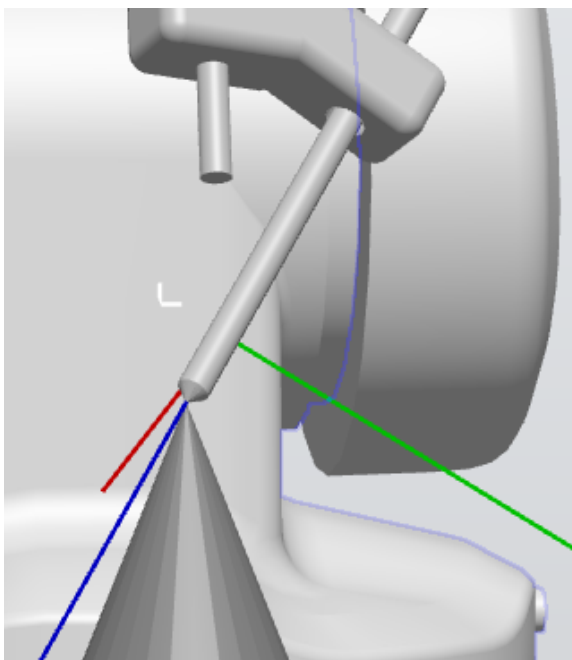
2.3.5. To start teaching TCP: Highlight the new TCP → Edit → Define

2.3.6. Select TCP&Z method

2.3.7. Select number of points: 3



2.3.8. Start Jogging the robot to the position that the pen tip touch the tip of the cone (using Front and Right view to see better)



2.3.9. Select the first position → Modify position to save the current position

2.3.10. Repeat with the second and third point but touching in different angles

2.3.11. For the Elongator point Z, align the pen with the Z axis in the simulation (pointing down)

→ move the robot, so the pen tip touch the tip of the cone, → move the robot in Z-axis

in the direction that the pen moving away from the cone.

2.3.12. Save the current position





Manual
Controller5 (KUNDAN-PC)

Motors On
Stopped (Speed 100%)



Program Data - tooldata - Define

Tool Frame Definition

Tool: TCP_Manual

Select a method, modify the positions and tap OK.

Method: TCP & Z No. of points: 3

Point	Status
Point 1	Modified
Point 2	-
Point 3	-
Elongator Point Z	-

Positions Modify Position OK Cancel

Production Window

Program Data

ROB_1
1/3

Manual
Controller5 (KUNDAN-PC)

Motors On
Stopped (Speed 100%)

Data of type: tooldata

Active filter:

Select the data you want to edit.

Scope: RAPID/T_ROB1

Change Scope

Name	Value	Module	
My_Mechanism_1	[TRUE,[[0,0,0],[1,0...	CalibData	Global
My_Mechanism_2	[TRUE,[[3.62455,-1...	CalibData	Global
TCP_Manual	[TRUE,[[0,0,0],[1,0...	Module1	Task
tool0	[TRUE,[[0,0,0],[1,0...	BASE	Global

New...

Edit

Refresh

View Data Types

Production Window

Program Data

Jogging

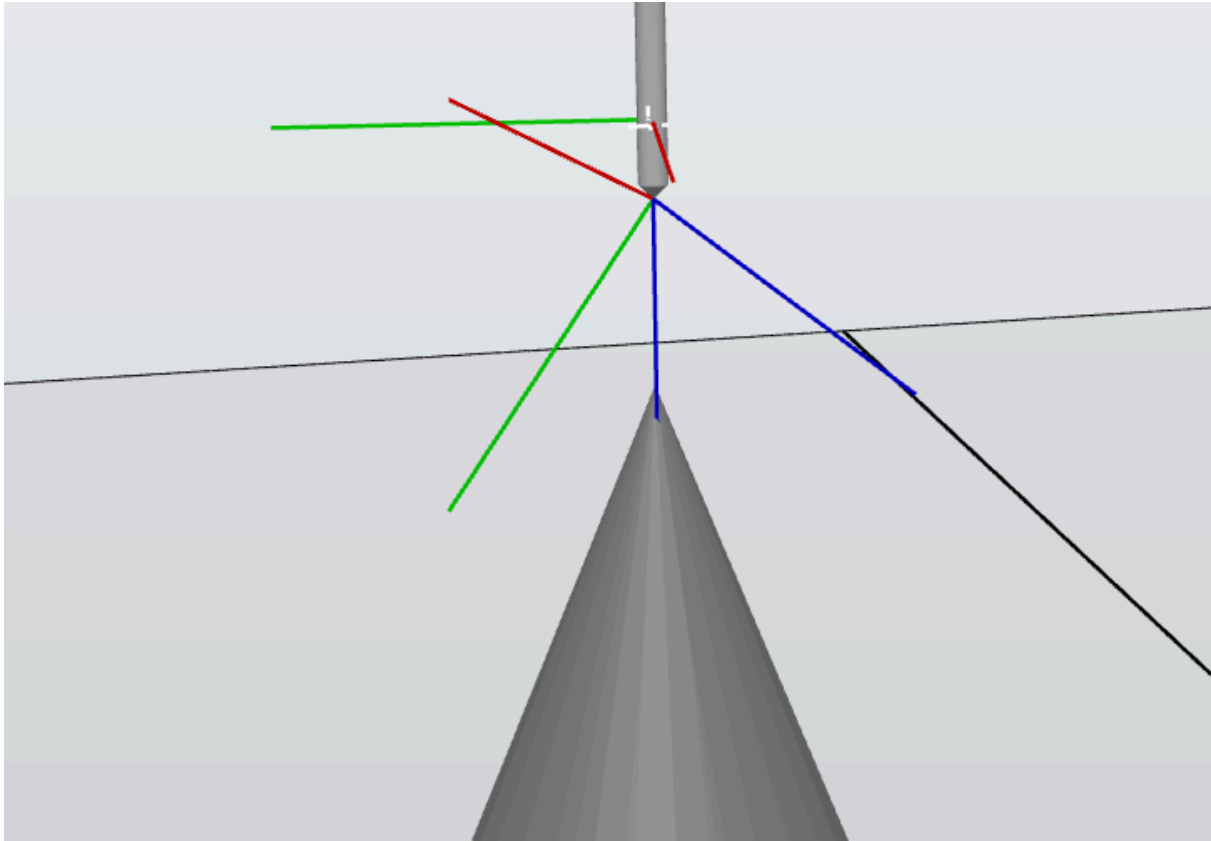
Program Data

ROB_1

2.3.14. Repeat step 1 but in the reverse direction (To station) → Tick on the new tool → OK

2.3.15. new TCP appear in the Tool section → select it

2.3.16. new TCP frame appear



3 Define robot working space- Practice

In this section, you will add a table and A4 paper to the simulation environment.

3.1 Define work object- table

3.1.1. Table dimension is $0.5 \times 0.25 \times 0.25\text{m}$

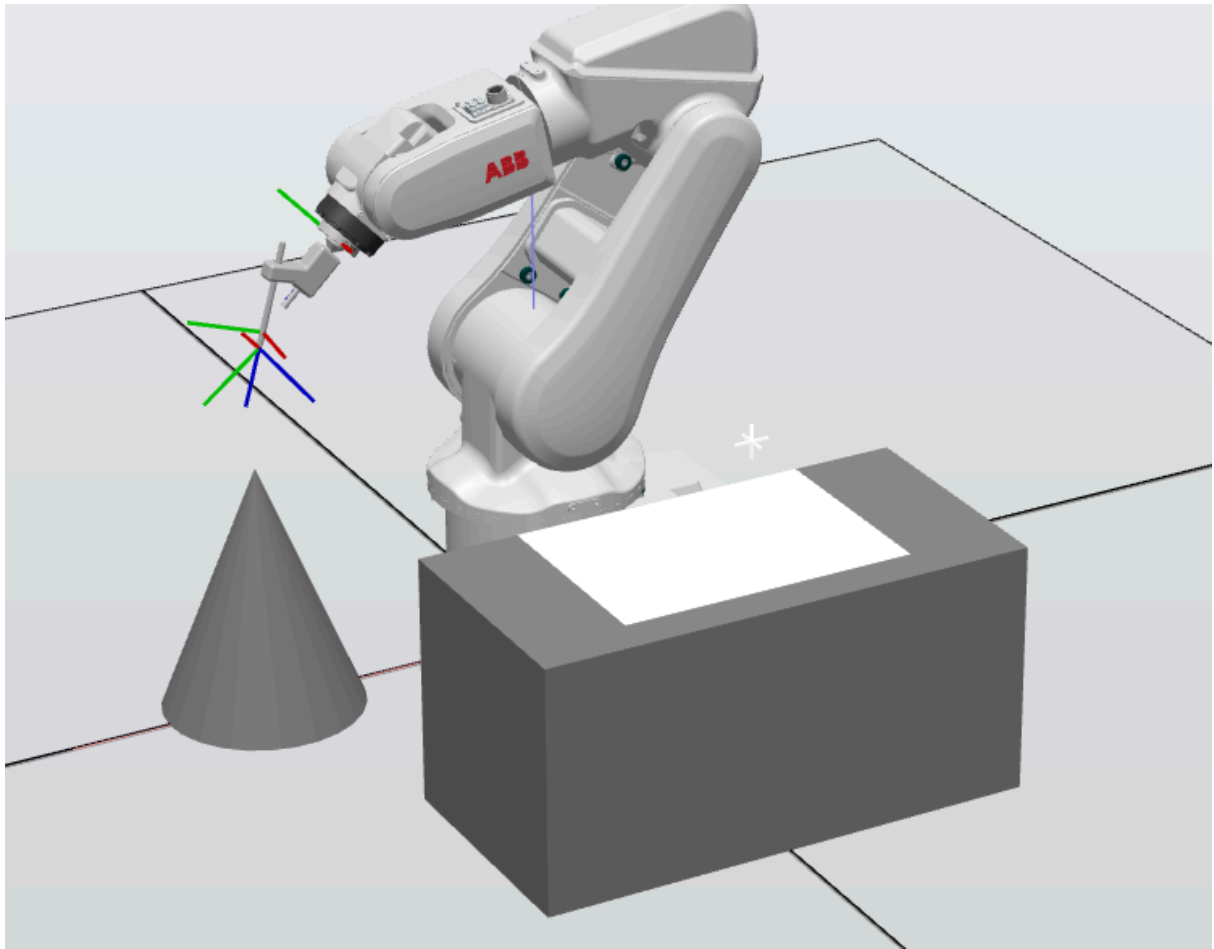
3.1.2. Table should be placed 0.25m away from the robot in X-axis

3.1.3. Table colour is black

3.2 Define work object- A4

3.2.1. Table dimension is $210 \times 297\text{ mm}$

3.2.2. Table should be placed on top of the table and on the edge facing toward the robot.



Week 4

Introduction to Robot Teaching mode and Programming mode for motion planning

1 Offline programming in the simulation

1.1 Target and Path

In this Exercise you will: Create a simple path for a pick-and-place task:

- Add a "PickUpPosition" target above an object.
 - Add a "PlaceDownPosition" target at a new location.
 - Create a path connecting these two targets.
 - Simulate to see the robot pick up the object and move it.

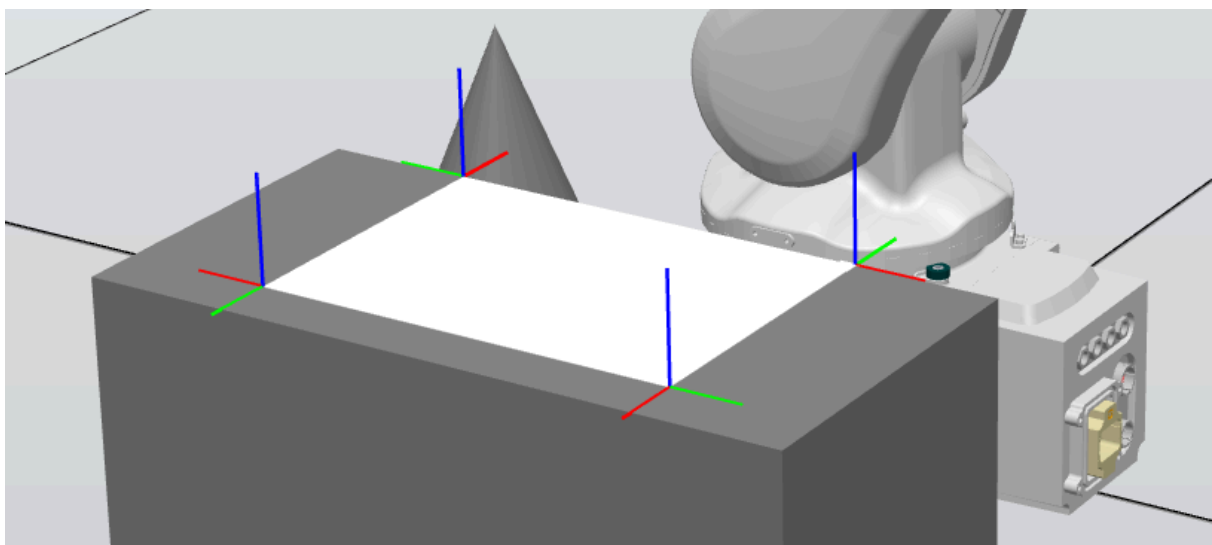
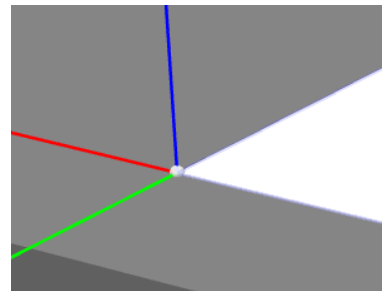
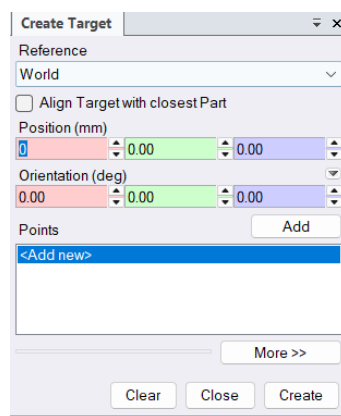
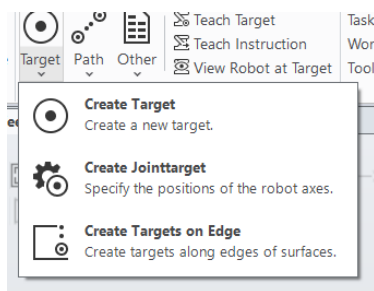
Prerequisites:

- Active RobotStudio Station: You need an open station in RobotStudio. If you don't have one, create a new station from the "File" menu.
- Active Robot and Controller: Create an IRB120 robot and controller if you don't have one. You can also use the previous week's simulation.

First, you need to create a few check point for the robot, which is called Targets.

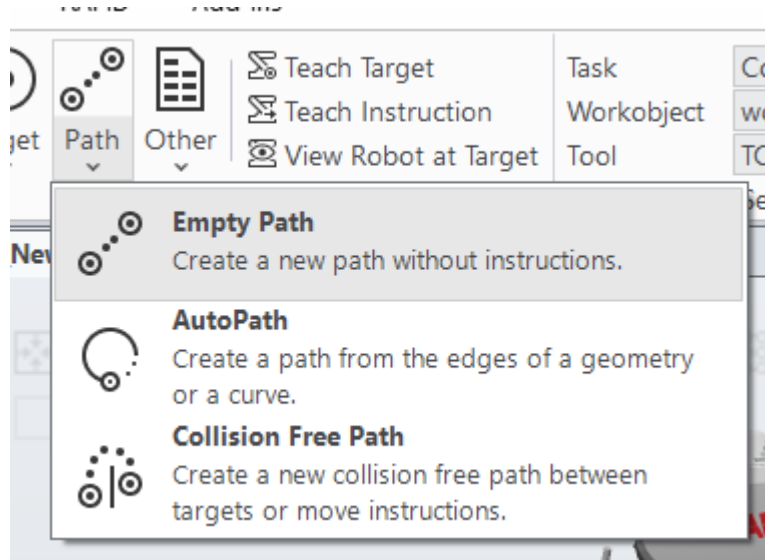
Follow step below to create Targets:

1. Target Toolbar: Locate the "Target" toolbar in RobotStudio (looks like a bullseye, on the top panel in Home tab).
2. Add Target: Click the small arrow → The list of different target placement method shows up



The you need to create Paths to guide your robot through targets. Following step below to create path:

1. Path Toolbar: Find the "Path" toolbar in RobotStudio (near the "Target" toolbar).
2. Add Path: Click the "Empty Path" button.

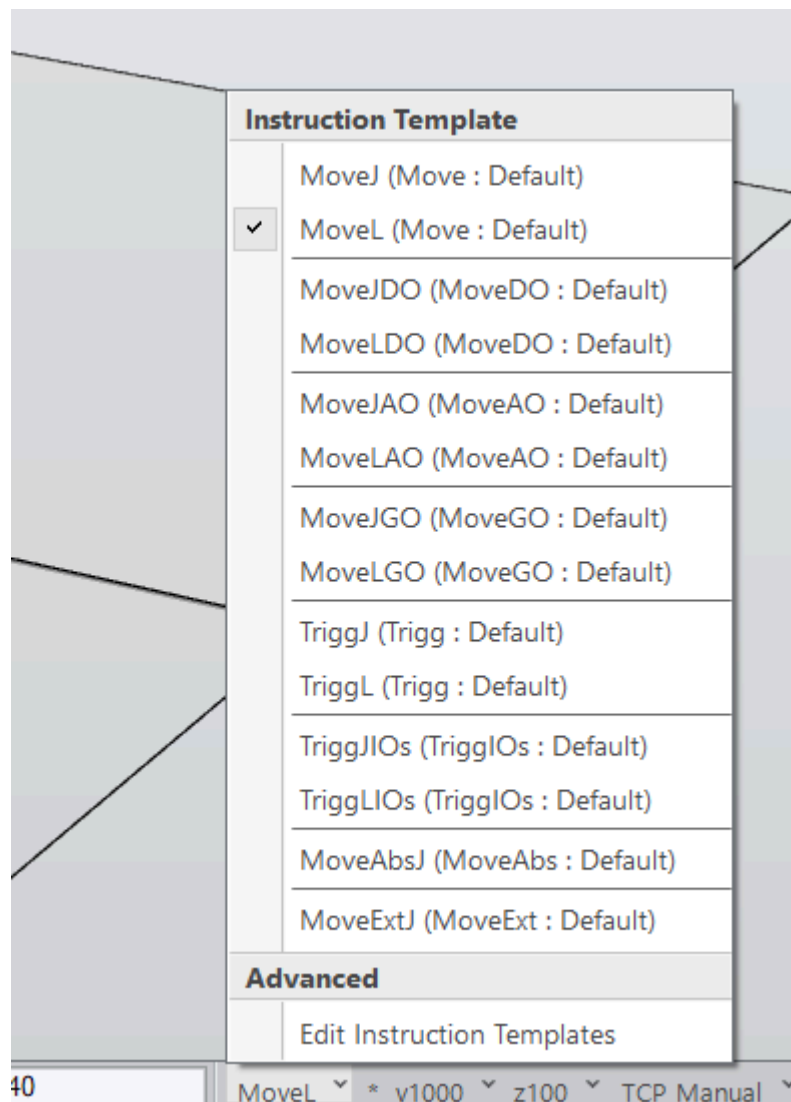


Adjust the Default Move instruction parameter: Find these parameters in the right bottom of your station (near to the controller status):

- Move Type: Choose the type of movement between targets (e.g., joint movement, linear movement), look into datasheet and table

Instruction	Purpose	Trajectory	Application	Use Cases
MoveAbsJ	Absolute joint movement	Joint interpolation	Precise control over joint positions	Fine-tuning robot positions, adjusting joint angles
MoveJ	Joint movement	Joint interpolation	Rapid movements between waypoints	Pick-and-place tasks, handling objects
MoveL	Linear movement	Straight-line path	Precise point-to-point movements	Conveyor tracking, following a moving conveyor
MoveC	Circular movement	Circular paths	Creating arcs or circular motions	Welding seams, smooth circular motions

Table 1.1: Summary of ABB Robot Move Instructions



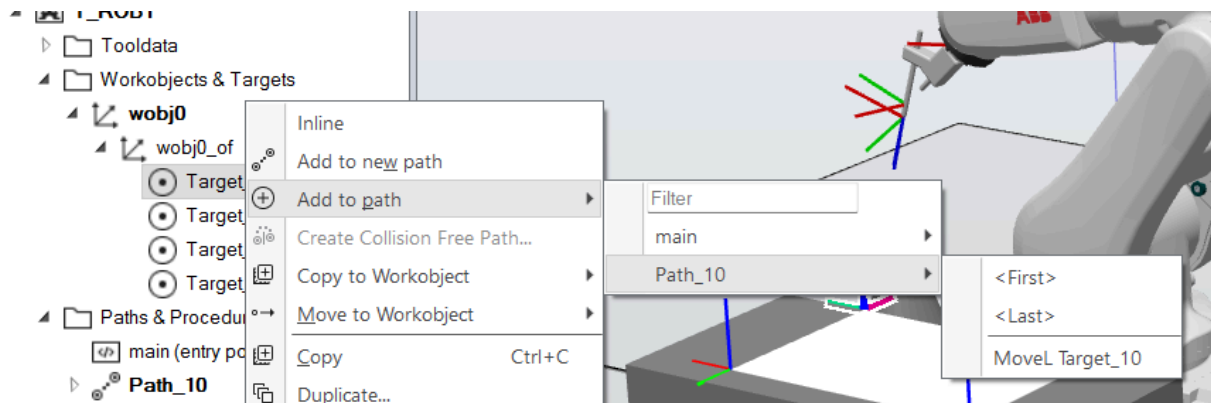
- Speed: Set the desired speed for each segment of the path.
- Zone Data: Adjust zone data for fine control over robot motion near targets.
 - Tool Data: Select your end effect/tool



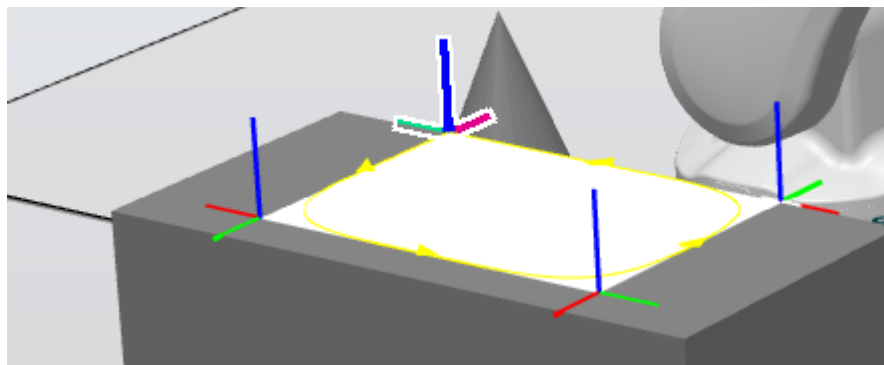
5. Add Targets to Path:

- Select your newly created target (one by one) in the tree view (on the left panel in Paths&Target tab).

- Right-click and choose "Add To Path"



- Repeat for each target in the desired order.



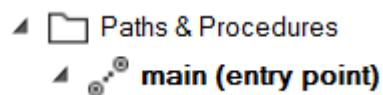
1.2 Path generator

once all the targets are added to a path.

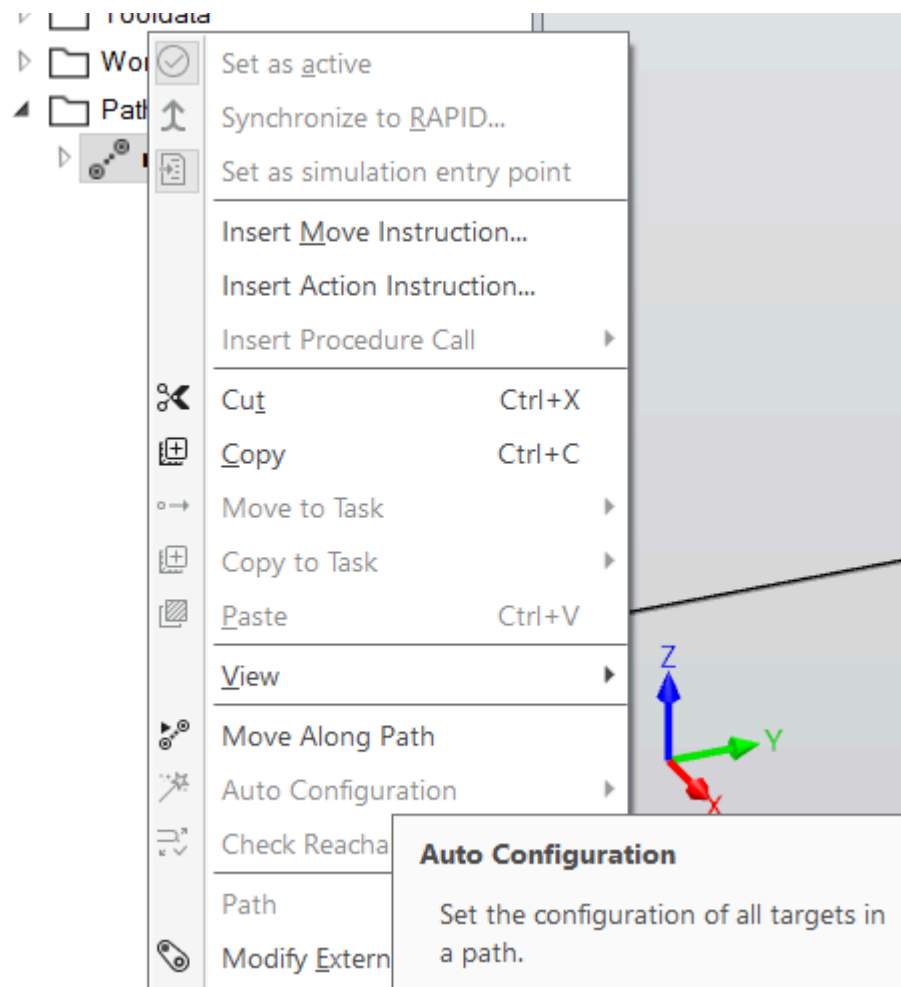
step -1

add all points to a path

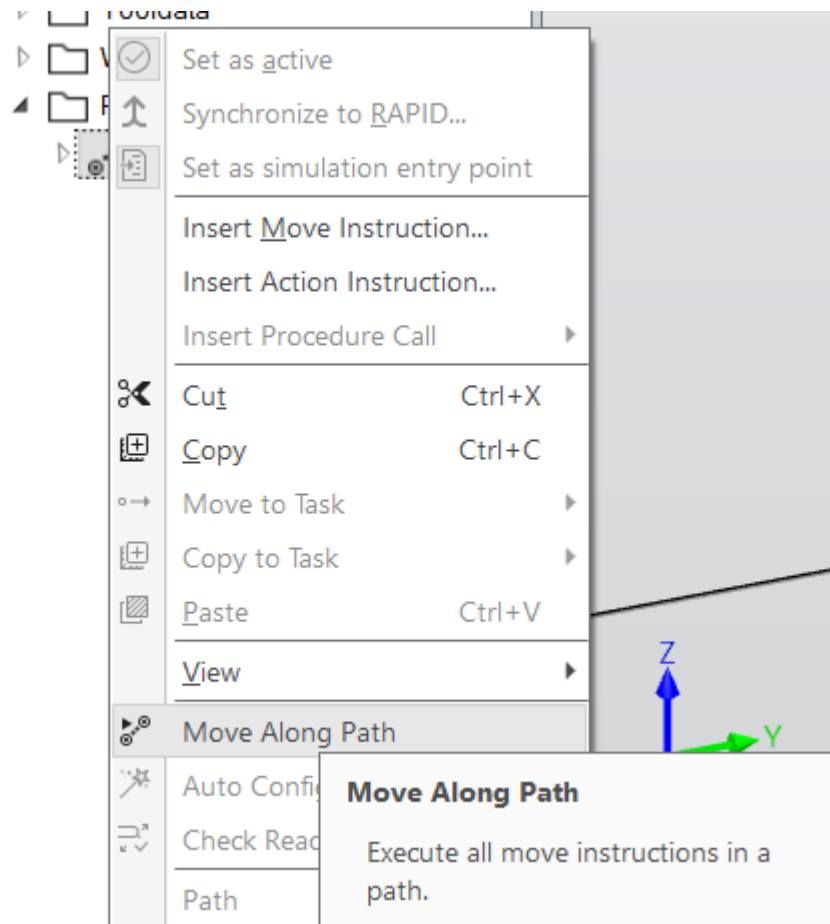
- rename the path to main so that it can take it as the starting point



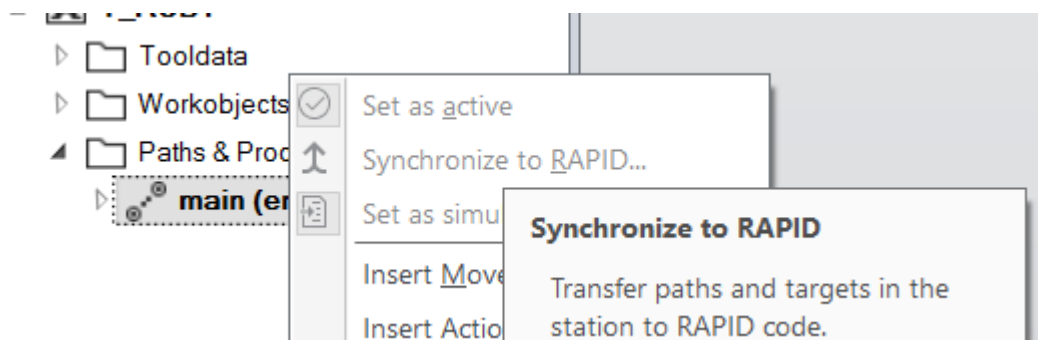
- go to auto config to verify and set the robot configuration for the path



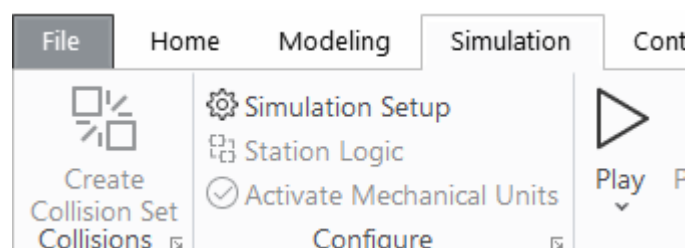
- if there is no error, then move on to the right click the path and try move along the path



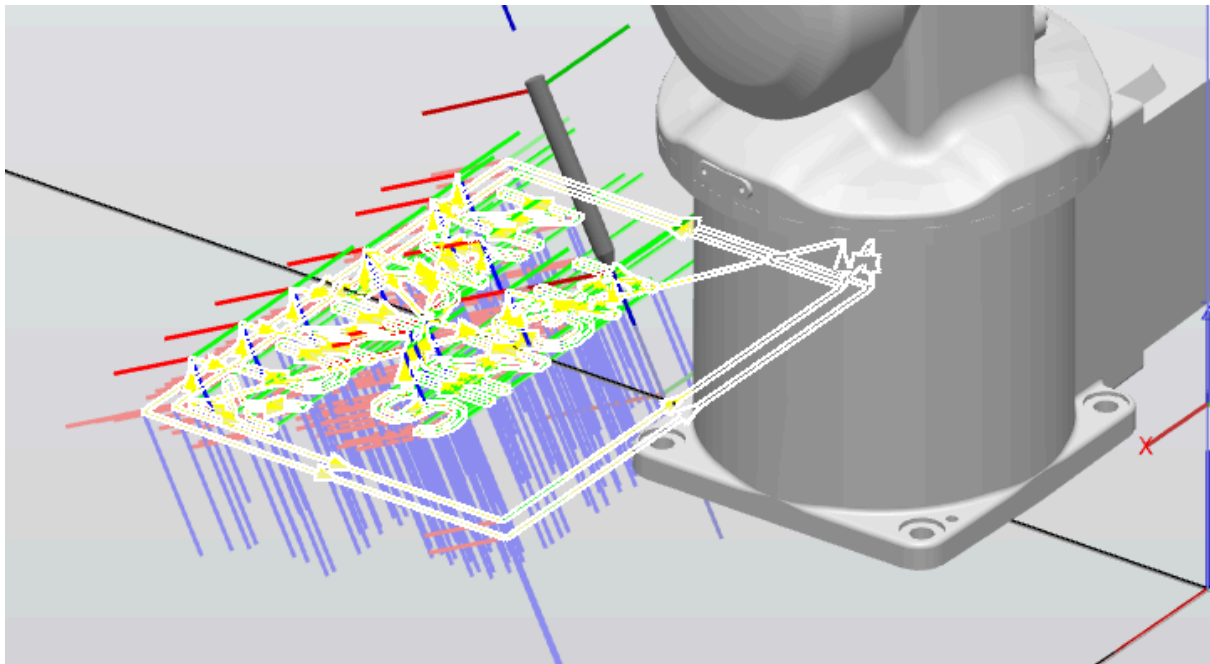
- once move along the path, it running smooth sync it to rapid



- now run the simulation.



Put your pen on an angle so that the orientation of your pen is same as writing with hand.

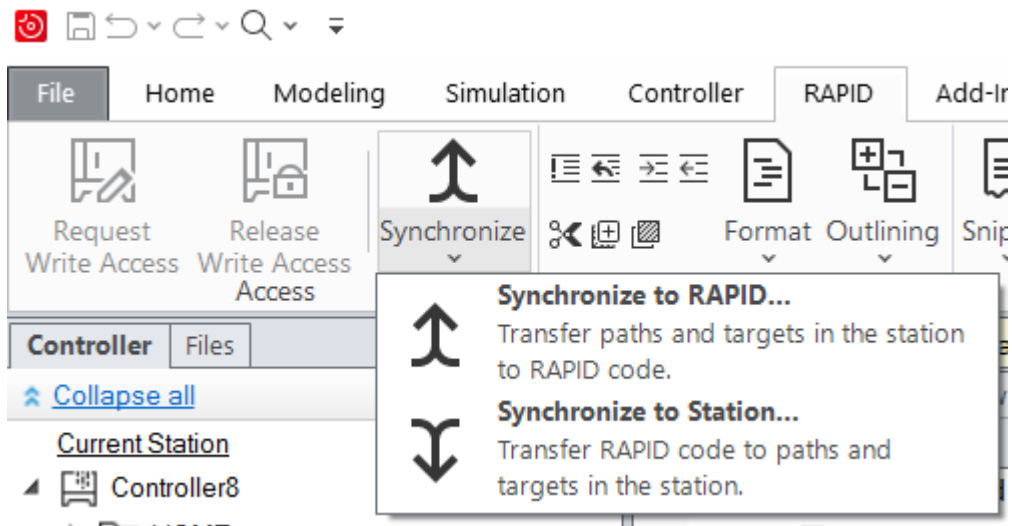


Set up the system the same as the actual lab setup

- The robot should be 16 mm above the base
- size of your paper and position as per the lab
- Choose a speed of v100 so that the robot writes at a natural speed, and it doesn't hamper with the outcome
- also, z should be fine because this is the accuracy of the path you will get.

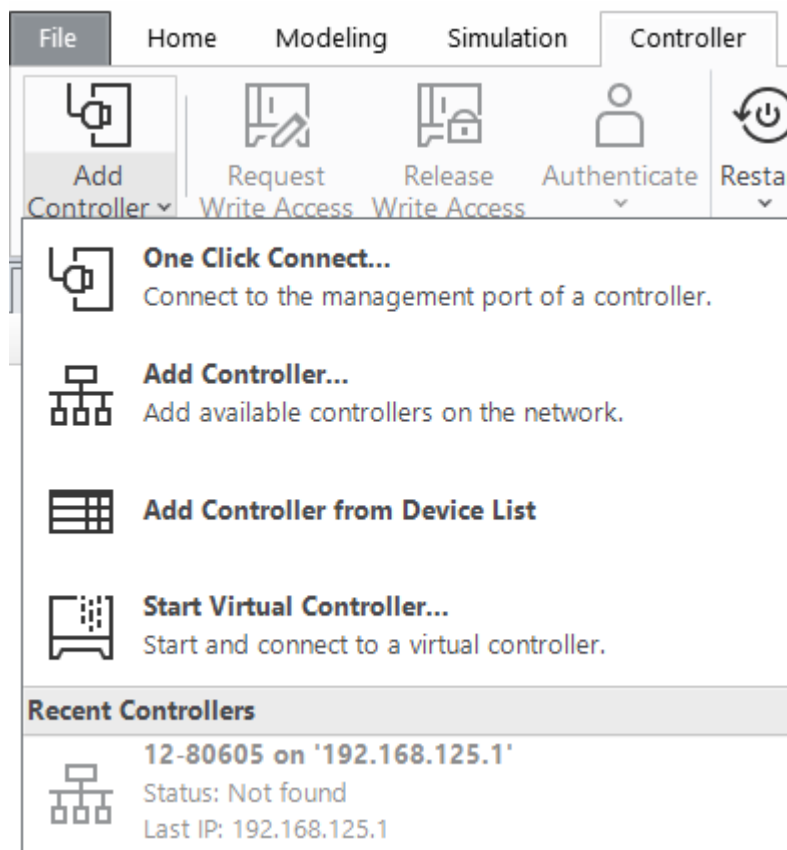
Once you are done with the simulation in your software:

- We need to transfer the program into the actual robot:
 - step - 1
 - Go to → Rapid → Sync to rapid
 -



Step -2

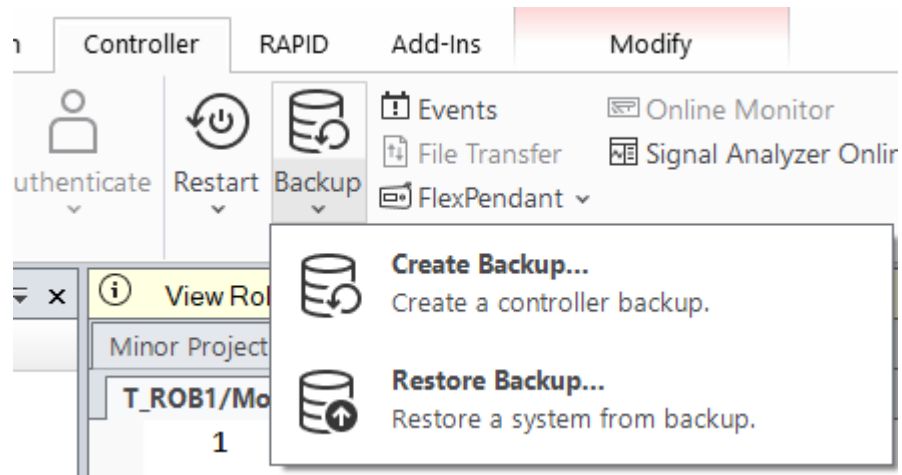
- Connect the robot to controller
- goto → Controller tab → click on add controller → click on one click connect (this will only happen if your bot is connected to the laptop)
-



- Login in as a default user

Step 3

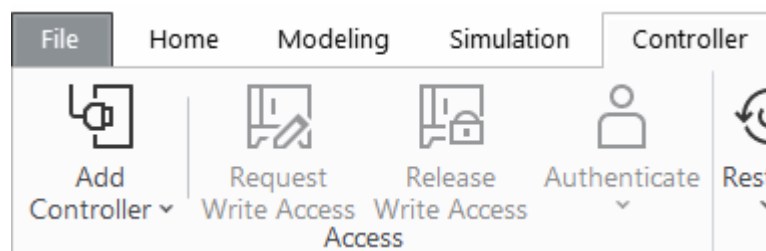
- Let's create a backup of it on your computer
- goto → Controller tab → click on back up
-



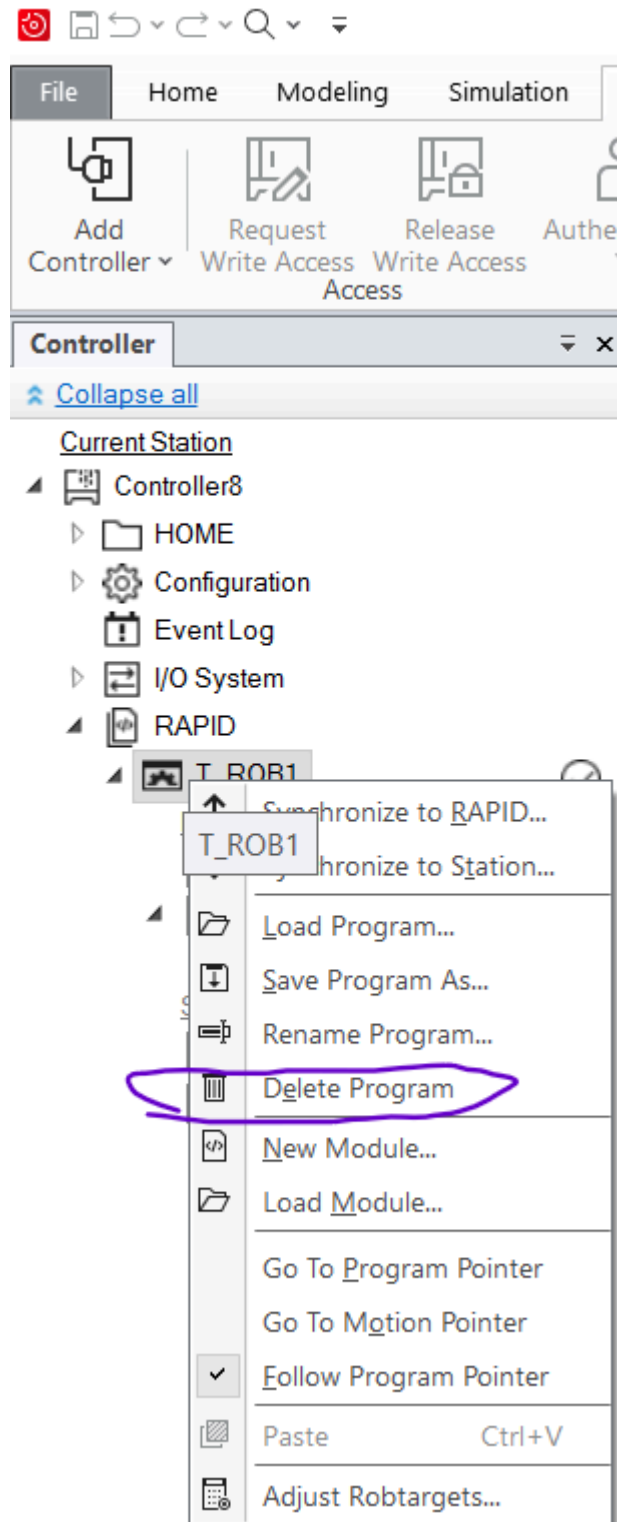
- follow the instructions for backup

step - 4

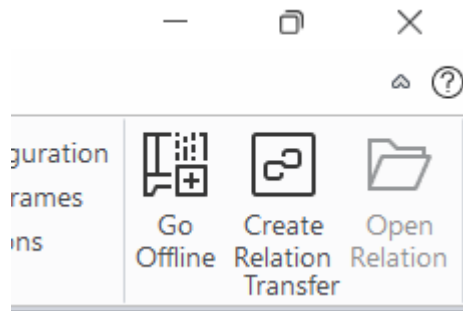
- Now you request a write access to the bot



- once you have the write access you need to delete existing program from the robot.
-



- Next you need to create a relation with robot
-



Create Relation ? X

Relation name:

First controller or backup

Second controller or backup

OK Cancel

- check the direction of transfer and select all the check boxes and click transfer now

Transfer

Target: 12-80605 (192.168.125.1) Change Direction Source: IRB120_3_58 (Station) **Transfer now...**

Transfer Configuration

Configure what items to transfer. The Action column will show a preview of the result. Everything in a mod

Name	Included	Action
WALL_DRAWING		
HOME	☒	
Data Declarations of Typ		
clock	☒	
num	☒	
System Modules		
user.sys	☒	No action (unchanged)
Files		
GSDML/GSDML-V2.3-AI	☒	Update
GSDML/GSDML-V2.32-f	☒	Update
GSDML/GSDML-V2.33-f	☒	Update
GSDML/GSDML-V2.34-f	☒	Update
T_ROB1	☐	
Data Declarations of Typ		
clock	☒	
num	☒	
robtarg	☒	
tooldata	☐	
wobjdata	☐	
Program Modules		
CalibData	☒	No action (unchanged)
Module1	☒	No action (unchanged)
System Modules		

Now you move on to the actual robot and play with the real controller
thanks